

Sparse convex hulls for network flows coupled to a simplex

Abstract

We study the convex hull of selected bilinear products between a bounded equality-constrained network flow and a simplex vector. The network and the retained product coordinates provide two distinct sources of structure: circulations factor across undirected blocks, and each block need retain only the simplex states observed there. We develop exact formulations and original-coordinate descriptions from these reductions, and examine the limits of small-coefficient descriptions. The starting point is classical simplex-vertex disaggregation; the structural question is how much of that formulation is needed for a sparse set of products.

1 Introduction

Network models often combine continuous routing decisions with a choice among mutually exclusive services or operating modes. A route-specific adjustment to the cost of a service gives a product of an arc flow and a service weight. Only a small fraction of the possible route-service pairs may occur in the model. Convexifying each product independently discards restrictions linking flows through the network and linking service weights through the simplex. An exact joint hull retains these restrictions, but its standard extended formulation introduces a copy of every arc flow for every simplex vertex.

This paper studies the structure of that joint hull when only selected products are retained. We distinguish the size of an extended formulation from the coefficients and separation cost of an original-coordinate description. We also distinguish equality-constrained flow models from network models with balance inequalities: adding slack arcs to convert the latter to the former can change the graph structure on which a specialized result depends. The elementary foundations below make the state and graph decompositions explicit, including degenerate capacities and zero simplex weights.

2 Model, prior work, and disaggregation

2.1 The sparse network-simplex hull

Let $D = (V, E)$ be a finite directed multigraph. Parallel arcs and self-loops are permitted. Its incidence matrix $A \in \{-1, 0, 1\}^{V \times E}$ has a -1 at the tail and a $+1$ at the head of a non-loop arc; a self-loop has a zero column. Thus $Ax = b$ uses the convention *incoming minus outgoing*. For $b \in \mathbb{Q}^V$ and finite $u \in \mathbb{Q}_+^E$, set

$$P = \{x \in \mathbb{R}^E : Ax = b, 0 \leq x \leq u\}. \quad (1)$$

Lower bounds can be handled by translating x and the corresponding product coordinates. Throughout, graph assumptions concern the graph in (1) after any such modeling transformations.

For an integer $m \geq 0$, write $[m] = \{1, \dots, m\}$ and

$$\Delta_m = \{y \in \mathbb{R}_+^m : \mathbf{1}^\top y \leq 1\}, \quad \lambda_j = y_j \ (j \in [m]), \quad \lambda_0 = 1 - \mathbf{1}^\top y. \quad (2)$$

The *states* are the $d = m + 1$ vertices $e^0 = 0, e^1, \dots, e^m$ of Δ_m ; e^j is the j th unit vector when $j \geq 1$. The residual state is state 0. An observation set $\mathcal{O} \subseteq E \times [m]$ specifies the products retained as original coordinates. We study

$$S = \{(x, y, z) : x \in P, y \in \Delta_m, z_{ej} = x_e y_j \quad ((e, j) \in \mathcal{O})\}, \quad (3)$$

$$H = \text{conv } S. \quad (4)$$

An unobserved product is absent from z , even if it is introduced temporarily in an extended formulation. State 0 has no observed coordinates. We use $\text{conv } \emptyset = \emptyset$. When $m = 0$, y and z are empty vectors and $H = P$.

The original coordinates are (x, y, z) . An extended formulation is a linear system in these coordinates and additional continuous variables whose projection is H . A separation procedure must either certify membership or produce a linear inequality valid for all of H and strictly violated by the candidate. Unless specified otherwise, a facet is taken relative to the affine hull. A bound on integer coefficients presupposes a stated normalization; a ratio of two nonzero coefficients is invariant under scaling the inequality. Arithmetic-operation bounds are distinguished from bounds on rational encoding length.

For comparison, the usual independent product relaxation adds, for every $(e, j) \in \mathcal{O}$, the four inequalities

$$0 \leq z_{ej}, \quad z_{ej} \leq u_e y_j, \quad z_{ej} \leq x_e, \quad z_{ej} \geq x_e - u_e(1 - y_j) \quad (5)$$

to $x \in P$ and $y \in \Delta_m$. These are the McCormick inequalities for the box $[0, u_e] \times [0, 1]$ [15]. They need not describe the joint hull in (4).

2.2 Relation to prior work

The general convexification principle is established. Disjunctive hull formulations [3] apply after expanding the simplex into its vertices. Simultaneous convexification over a polytope times a simplex is developed by Davarnia et al. [5]. Khademnia and Davarnia [12] specialize this setting to network polytopes: their electronic companion, p. 1, equation (25), gives a full extended hull,¹ and their aggregation-and-projection framework gives a complete original-space description in principle. Their explicit tree and forest constructions identify more structured inequality families; the forest recipe for multiple simplex coordinates should not be confused with the entire projection framework. Their general framework permits balance inequalities, and their Section 3 represents equality balances by pairs of opposite inequalities. Thus their network specialization includes (1); here we exploit its underlying equality circulation space. We use the published article and its electronic companion for equation and example numbering.

The shared-simplex gluing principle is also known. In particular, Davarnia [4, Proposition 2.6] proves that convexifying Cartesian components separately is exact when the components share the same simplex. The cited locator belongs to the dissertation, which is distinct from the 2017 journal article. Our block decomposition applies this principle to the independent circulation spaces of an equality-constrained network. We give a direct proof to make the treatment of sparse observations and zero weights explicit.

Several established methods explain the subsequent structural reductions. Reduced reformulation–linearization uses multiplied linear equations to remove common-factor products [14]. In networks, the remaining freedom can be identified with circulations supported on unobserved arcs. The network representation of Cayley hulls of Kis and Horváth [13] gives cut descriptions and transportation

¹The published electronic companion is available at [this direct supplement link](#).

projections; their Section 5.7 is a close predecessor for the parallel-path construction. The support geometry of Minkowski sums and edge-generated zonotope refinements is classical [10, 17]. These are ingredients of the graph-specific formulations and separation procedures, rather than new general convexification mechanisms.

Two further comparisons delimit coefficient obstructions. Three-way transportation universality is established by De Loera and Onn [6], and large coefficients in general 0/1 polytopes are classical [2]. A restricted network realization must therefore be specified as part of a coefficient result. Moreover, the integrality of *aggregate* flows on two-terminal series-parallel networks does not give a hull description in selected state-flow coordinates. Almoghrabi et al. [1, Remark 1] explicitly distinguish aggregate integrality from decomposability of the vector of individual commodity flows. Our observed products retain such individual-state information. Neither a sparse extended formulation nor polynomial separation precludes large numerical coefficients after projection.

2.3 The known exact disaggregation

For $\lambda \geq 0$, define the scaled flow system

$$P(\lambda) = \{f \in \mathbb{R}^E : Af = \lambda b, 0 \leq f \leq \lambda u\}. \quad (6)$$

For $\lambda > 0$, this is λP ; at $\lambda = 0$ it is $\{0\}$, including when P is empty. Using (6) avoids assigning an ambiguous meaning to $0\emptyset$.

Proposition 2.1 (Simplex-vertex disaggregation). *A point (x, y, z) belongs to H if and only if $y \in \Delta_m$ and there exist state flows $f^0, \dots, f^m \in \mathbb{R}^E$ such that*

$$f^k \in P(\lambda_k) \quad (k \in \{0, \dots, m\}), \quad \sum_{k=0}^m f^k = x, \quad f_e^j = z_{ej} \quad ((e, j) \in \mathcal{O}). \quad (7)$$

The equivalence holds even if P is empty. If the system is feasible, it provides a convex decomposition into at most $m + 1$ points of S .

Proof. For a point of S , set $f^k = \lambda_k x$. This satisfies all displayed relations, since $\sum_k \lambda_k = 1$. The system is linear in (x, y, z, f) , so its projection is convex and contains H .

Conversely, suppose the system is feasible. If $\lambda_k = 0$, the finite upper bounds give $f^k = 0$. For each positive weight let $x^k = f^k / \lambda_k \in P$ and define the graph point $p^k = (x^k, e^k, z^k)$ by $z_{ej}^k = x_e^k(e^k)_j$. Its weighted average has x coordinate $\sum_k f^k = x$, y coordinate $\sum_k \lambda_k e^k = y$, and observed product coordinate $\lambda_j x_e^j = f_e^j = z_{ej}$ when $\lambda_j > 0$. If $\lambda_j = 0$, that observed coordinate is zero on both sides. Thus $(x, y, z) = \sum_{k: \lambda_k > 0} \lambda_k p^k \in H$. There is at least one positive weight, so feasibility also implies $P \neq \emptyset$. This proves the empty-set assertion. $\square$

The formulation has $(m + 1)|E|$ state-flow variables before elimination. Its rational size is polynomial in the ordinary network and simplex input, so generic polynomial-time hull optimization and separation already follow from linear programming. The structural issue is avoiding this full state-by-arc expansion, or projecting it into useful original-coordinate inequalities. In particular, eliminating $f^0 = x - \sum_{j=1}^m f^j$ keeps the residual bounds

$$0 \leq x - \sum_{j=1}^m f^j \leq \left(1 - \sum_{j=1}^m y_j\right)u. \quad (8)$$

These bounds cannot in general be discarded.

3 Independent blocks and locally merged states

3.1 Circulation factorization

All graph connectivity terminology below refers to the undirected multigraph underlying D . Isolated vertices impose only their balance equation. Let E_{br} be the bridges. Let $\mathcal{B}$ be the cyclic blocks: the maximal biconnected edge blocks that contain a cycle, with each self-loop treated as its own block. A pair of parallel edges can form a cycle of length two. The sets E_{br} and E_B for $B \in \mathcal{B}$ partition E . The cycle rank of a connected block is

$$r_B = |E_B| - |V_B| + 1. \quad (9)$$

In particular, a loop has rank one. Rank refers to the ambient circulation space, even when capacities reduce the dimension of the feasible polytope.

The equation $Av = b$ is solvable exactly when the entries of b sum to zero on every connected component. Necessity follows by summing the incidence rows in a component. For sufficiency, choose a spanning forest, set the nonforest coordinates to zero, and determine each tree-edge flow by the balance sum on one side of its fundamental cut, with the sign determined by the arc orientation. This satisfies every balance equation. This construction uses $O(|V| + |E|)$ arithmetic operations, and for rational b produces a rational v of polynomial encoding length. The reference flow v need not obey its capacity bounds. If the component condition fails, P and H are empty.

Fix such a reference vector v . For each B , let A_B be the incidence matrix restricted to columns in E_B and rows in V_B , and define

$$K_B = \{h \in \mathbb{R}^{E_B} : A_B h = 0, -v_{E_B} \leq h \leq u_{E_B} - v_{E_B}\}. \quad (10)$$

Unlike P , this is a domain of circulation *deviations*; it need not contain the origin.

Lemma 3.1 (Block factorization). *Every $g \in \ker A$ vanishes on E_{br} and restricts to a circulation on each cyclic block. Conversely, combining arbitrary block circulations and zero bridge coordinates gives an element of $\ker A$. Consequently, if every bridge satisfies $0 \leq v_e \leq u_e$, the map*

$$x_e = v_e \quad (e \in E_{\text{br}}), \quad x_{E_B} = v_{E_B} + h_B \quad (B \in \mathcal{B}) \quad (11)$$

is an affine bijection from $\prod_{B \in \mathcal{B}} K_B$ to P . If a bridge bound fails, P is empty. The product of an empty family of block domains is a singleton.

Proof. Choose a spanning forest of the underlying multigraph, omitting loops. Each nonforest edge gives a signed fundamental-cycle circulation: use the unique forest path between its ends, or just the edge itself for a loop. These vectors form a basis of $\ker A$. Indeed, subtracting from a circulation the combination that cancels all nonforest coordinates leaves a circulation supported on a forest, which is zero by successive leaf elimination.

Each fundamental cycle lies in a single cyclic block. Hence its vector is zero on every bridge, and on every block it is either zero or a circulation of that block. The basis expansion proves the first assertion. For the converse, extend each block circulation by zero to E . Its incidence is zero at every vertex, including articulation vertices, so their sum is a global circulation. The sum is direct because the block edge sets are disjoint. Applying this decomposition to $x - v$ proves that $Ax = b$ is equivalent to the circulation conditions and the fixed bridge coordinates in (11). The bounds then separate by edge block, giving precisely (10) and the bridge conditions. $\square$

In particular, an articulation vertex does not carry an additional coupling variable between block deviations. Its fixed net balance is already carried by v ; every block deviation has zero net balance at that vertex separately. Any spanning tree in B gives a fundamental-cycle matrix $C_B \in \{0, \pm 1\}^{E_B \times r_B}$ with chord rows equal to the identity. Thus $h_B = C_B t_B$ gives unique cycle coordinates $t_B \in \mathbb{R}^{r_B}$. These coordinates remain useful when K_B is lower-dimensional.

3.2 The state merger and its exact refinement

For a nonempty convex set K , use the usual scalar multiplication $\alpha K = \{\alpha h : h \in K\}$, so $0K = \{0\}$. For nonnegative scalars,

$$\alpha_1 K + \cdots + \alpha_q K = (\alpha_1 + \cdots + \alpha_q) K. \quad (12)$$

If the total weight is positive, the inclusion from left to right follows by dividing by that weight and taking a convex combination; the reverse inclusion uses the same point of K in each summand. If the total is zero, both sides are $\{0\}$. An empty sum is also interpreted as $\{0\}$.

For each block define its observed labels and their count by

$$J_B = \{j \in [m] : (e, j) \in \mathcal{O} \text{ for some } e \in E_B\}, \quad a_B = |J_B|. \quad (13)$$

This is a structural definition: a label in J_B remains present when $y_j = 0$. Let $*_B$ denote a local merged state, distinct from the global residual state 0, and set

$$I_B = J_B \cup \{*_B\}, \quad \lambda_{B,j} = y_j \ (j \in J_B), \quad \lambda_{B,*} = 1 - \sum_{j \in J_B} y_j. \quad (14)$$

The merged state collects all labels absent from J_B and state 0. Different blocks may merge different global states.

Theorem 3.2 (Exact blockwise state reduction). *Assume $P \neq \emptyset$. A point (x, y, z) belongs to H if and only if $x \in P$, $y \in \Delta_m$, every observed bridge satisfies*

$$z_{ej} = v_e y_j \quad (e \in E_{\text{br}}, (e, j) \in \mathcal{O}), \quad (15)$$

and for every cyclic block B there exist vectors $h^{B,k} \in \mathbb{R}^{E_B}$, $k \in I_B$, satisfying

$$h^{B,k} \in \lambda_{B,k} K_B \quad (k \in I_B), \quad (16)$$

$$\sum_{k \in I_B} h^{B,k} = x_{E_B} - v_{E_B}, \quad (17)$$

$$h_e^{B,j} = z_{ej} - y_j v_e \quad ((e, j) \in \mathcal{O}, e \in E_B). \quad (18)$$

The block conditions yield state flows in (7), and hence a decomposition into at most $m + 1$ graph points, by explicit proportional refinement.

Proof. By lemma 3.1, every K_B is nonempty. Start with state flows in (7). Their deviations $g^k = f^k - \lambda_k v$ are circulations. On B , they satisfy $g_{E_B}^k \in \lambda_k K_B$: division proves this for positive weights, and both deviations and scaled domains are zero for zero weights. For $j \in J_B$ take $h^{B,j} = g_{E_B}^j$, and sum the other deviations into $h^{B,*}$. Identity (12) proves (16). Summing the deviations proves (17), and the observed coordinates give (18). On bridges the deviations vanish, proving (15).

Conversely, fix feasible local vectors. For each global state k and block B , if $k \in J_B$, set $g_{E_B}^k = h^{B,k}$. For the remaining states use

$$g_{E_B}^k = \begin{cases} (\lambda_k/\lambda_{B,*})h^{B,*}, & \lambda_{B,*} > 0, \\ 0, & \lambda_{B,*} = 0. \end{cases} \quad (19)$$

If the merged weight is zero, every constituent weight is zero as well. In every case $g_{E_B}^k \in \lambda_k K_B$, and the refined vectors sum to the block deviation. Put $g_e^k = 0$ on bridges and combine blocks. The block factorization gives $Ag^k = 0$. Then $f^k = \lambda_k v + g^k$ satisfies $Af^k = \lambda_k b$. Its capacity bounds hold on blocks by the scaled domains, and on bridges because $P \neq \emptyset$ implies $0 \leq v_e \leq u_e$. Equations (17) and (18), together with the bridge equations, give $\sum_k f^k = x$ and all observed products. Apply [proposition 2.1](#). $\square$

The domain condition (16) is itself linear:

$$A_B h^{B,k} = 0, \quad -\lambda_{B,k} v_{E_B} \leq h^{B,k} \leq \lambda_{B,k} (u_{E_B} - v_{E_B}). \quad (20)$$

Finite bounds force $h^{B,k} = 0$ at zero weight. No division by a variable is needed in the formulation; division appears only in the recovery proof at strictly positive weights. If $J_B = \emptyset$, the block has one merged state of weight one and its conditions reduce to $x_{E_B} - v_{E_B} \in K_B$, already enforced by $x \in P$. Thus unobserved blocks require no new variables or constraints.

The theorem is a constructive specialization of the known shared-simplex gluing principle. It does not introduce independent simplex vectors for different blocks: all use the same global y . It also does not merge observed state slices, whose domains can differ after the products are fixed.

Example 3.3 (A common matrix is not sufficient for merging). Let $M = (-1, 1, 2)^\top$, $c^1 = (0, 1, 100)^\top$, and $c^2 = (0, 100, 2)^\top$. Both polytopes $Q_i = \{q \in \mathbb{R} : Mq \leq c^i\}$ equal $[0, 1]$. For weights $\alpha_1 = \alpha_2 = 1/2$, the weighted Minkowski sum is still $[0, 1]$, whereas the aggregated rows $Mq \leq (c^1 + c^2)/2$ allow $q = 2$. The two upper bounds become $q \leq 101/2$ and $2q \leq 51$. Thus replacing state-specific systems by the sum of their right-hand sides is generally inexact, even for a common matrix and nonempty bounded domains. This is the standard reaggregation boundary discussed by Kis and Horváth [13, Section 2, equation (7)]. The sets in this example are identical, so the set identity (12) still holds. What fails is its replacement by addition of the different, redundant right-hand-side descriptions. In [theorem 3.2](#), every state instead uses the single fixed representation (10), with its bounds scaled by the state weight as in (20). Adding these scaled bounds gives that same representation scaled by the total weight, so the set identity justifies the merged formulation.

Finally, these results concern the hull of (3). Intersecting H with additional linear side constraints yields a valid relaxation of the graph with those constraints, but need not give its exact convex hull. Such constraints can couple blocks or alter conditional state domains; exactness then requires a separate argument.

4 Compression by cycles and observations

This section compresses the exact block formulation in two steps. First, cycle coordinates replace arc coordinates and suppress long paths. Second, observations determine some cycle coordinates, leaving only cycles supported on unobserved arcs. We assume $P \neq \emptyset$; if the original flow domain is empty, its hull is described by any inconsistent linear system.

4.1 Fixed coordinates and reduced graphs

The structural parameters can be applied after elementary exact preprocessing. This is useful when capacities force a nominal cycle to carry no freedom.

Lemma 4.1 (Deleting fixed arcs). *Let $F \subseteq E$ and suppose $x_e = c_e$ throughout P for every $e \in F$, where $c \in \mathbb{Q}^F$ is known. Set $R = E \setminus F$, $b' = b - A_{:,F}c$, and*

$$P' = \{x_R : A_{:,R}x_R = b', 0 \leq x_R \leq u_R\}, \quad \mathcal{O}' = \mathcal{O} \cap (R \times [m]).$$

The hull for $(P', \mathcal{O}')$, together with $x_F = c$ and $z_{ej} = c_e y_j$ for observed $e \in F$, is affinely isomorphic to H . If F contains every coordinate fixed throughout P , then

$$\text{aff } P' = \{x_R : A_{:,R}x_R = b'\}. \quad (21)$$

Proof. Feasible vectors of P restrict to P' . Conversely, adjoining c to a vector of P' satisfies the original balances and bounds, because $0 \leq c \leq u_F$. The same bijection at graph points adjoins the affine coordinates $z_{ej} = c_e y_j$, so it commutes with convexification.

For the last assertion, every remaining coordinate is nonconstant on P' . For each $e \in R$, choose two feasible vectors differing in coordinate e ; their midpoint has $0 < x_e < u_e$. Average these midpoints over $e \in R$. The resulting feasible vector is strictly inside every remaining bound. A sufficiently small relative neighborhood of that vector in the balance affine space therefore remains feasible, proving (21). If R is empty, both spaces are the same singleton and the assertion is immediate. $\square$

The lemma takes the fixed coordinates as input; it does not require a particular detection algorithm. Deleting them can split blocks or reduce their cycle ranks. Once all fixed coordinates are removed, the remaining cycle space is the direction space of the feasible affine hull. Conditional slices at a particular observation or zero state weight can still have smaller dimension.

4.2 Suppressing paths

Fix a cyclic block B . Degrees in this subsection are degrees *within the block*, with loops counted twice. For $r_B \geq 2$, suppress maximal paths whose internal vertices have degree two, and give each resulting core edge an arbitrary orientation. A rank-one block is a single cycle; represent its entire cyclic traversal by one core loop. This includes two parallel edges and an original self-loop.

Write $\mathcal{P}_B$ for the core edges, or equivalently the original paths, and $k_B = |\mathcal{P}_B|$. For an original arc e on path p , let $\varepsilon_e \in \{-1, 1\}$ compare its orientation with the chosen path traversal. A circulation has constant signed deviation on a path:

$$x_e = v_e + \varepsilon_e \delta_p \quad (e \in p). \quad (22)$$

The reason is balance of the block circulation at each internal degree-two vertex. Balances at the unsuppressed vertices say precisely that δ is a circulation of the core. Arc bounds give the path interval

$$\begin{aligned} L_p &= \max_{e \in p} \ell_e, & U_p &= \min_{e \in p} u'_e, \\ [\ell_e, u'_e] &= \begin{cases} [-v_e, u_e - v_e], & \varepsilon_e = 1, \\ [v_e - u_e, v_e], & \varepsilon_e = -1. \end{cases} \end{aligned} \quad (23)$$

All intervals are finite. Their joint feasibility follows from $P \neq \emptyset$.

Suppression preserves cycle rank. If $r_B \geq 2$, the connected core has minimum degree at least three: otherwise the original block would have been a single cycle. If its vertex count is n_B , then $2k_B \geq 3n_B$ and $r_B = k_B - n_B + 1$, so

$$n_B \leq 2r_B - 2, \quad k_B \leq 3r_B - 3. \quad (24)$$

For rank one, $k_B = 1$. In particular, $k_B \leq 3r_B$ in every case.

Choose a spanning tree of the core and let $\hat{C}_B \in \{0, \pm 1\}^{k_B \times r_B}$ be its fundamental-cycle matrix, normalized to have identity chord rows. Its columns parametrize core circulations bijectively. Thus the feasible core domain is

$$T_B = \{t \in \mathbb{R}^{r_B} : L_p \leq (\hat{C}_B t)_p \leq U_p \quad (p \in \mathcal{P}_B)\}. \quad (25)$$

It is nonempty and bounded, including in lower-dimensional cases. Boundedness follows already from the chord rows. For later use, choose one original representative arc $e(p)$ on each path and put

$$s_p(x) = \varepsilon_{e(p)}(x_{e(p)} - v_{e(p)}), \quad q_{ej} = \varepsilon_e(z_{ej} - v_e y_j). \quad (26)$$

For $x \in P$, $s_p(x)$ is its core deviation. A separate observation q_{ej} is retained for *every* observed original arc, including multiple observations of one path.

Theorem 4.2 (Cycle-coordinate extended hull). *The hull H has an exact rational extended formulation with*

$$\sum_{B \in \mathcal{B}} r_B a_B \quad (27)$$

additional variables and

$$O \left(|V| + |E| + m + |\mathcal{O}| + \sum_{B: a_B > 0} r_B (a_B + 1) \right) \quad (28)$$

rows, counting the original domain constraints. Coefficients of x , z , and additional variables can all be chosen from $\{0, \pm 1\}$. Coefficients of y and the right-hand sides may depend on the rational network data. All coefficient encoding lengths are polynomial in the input length.

Proof. Retain $x \in P$, $y \in \Delta_m$, and the bridge equations (15). For each B with $a_B > 0$, introduce $t^{B,j} \in \mathbb{R}^{r_B}$ for $j \in J_B$ and impose

$$y_j L_p \leq (\hat{C}_B t^{B,j})_p \leq y_j U_p, \quad j \in J_B, p \in \mathcal{P}_B, \quad (29)$$

$$q_{ej} = (\hat{C}_B t^{B,j})_p, \quad (e, j) \in \mathcal{O}, e \in p \subseteq E_B, \quad (30)$$

$$\lambda_{B,*} L_p \leq s_p(x) - \sum_{j \in J_B} (\hat{C}_B t^{B,j})_p \leq \lambda_{B,*} U_p, \quad p \in \mathcal{P}_B. \quad (31)$$

The aggregate deviation $s(x)$ is a core circulation. Hence the middle quantity in (31) is a core circulation as well. These constraints are exactly [theorem 3.2](#) in core coordinates, with the merged state eliminated by its sum equation. At a zero weight, the finite path bounds force every core path value to zero; the identity chord rows force the entire state vector to zero. Consequently the equivalence includes all simplex boundary strata. Blocks with $a_B = 0$ add no constraints, as in the block theorem.

There are $r_B a_B$ variables, $2k_B(a_B + 1)$ bound rows, and one equation per block observation. Bound (24) proves the row count. Every state variable has coefficient 0, 1, or -1 . An observation row has one original product coordinate, and each residual row uses one original flow coordinate through $s_p(x)$; no sums of different cycle expressions for that flow coordinate are necessary. This proves the stated coefficient bound. Reference flows are sums of rational balances; path endpoints are selected signed capacity offsets. The remaining coefficients are sums of these data, with polynomial rational encoding length. $\square$

4.3 Eliminating observed directions

Define the observed and unobserved arcs for a block and label by

$$O_{Bj} = \{e \in E_B : (e, j) \in \mathcal{O}\}, \quad U_{Bj} = E_B \setminus O_{Bj}.$$

Let $c(V_B, U_{Bj})$ count all connected components of this unobserved undirected subgraph, including isolated vertices. Its cycle rank is

$$\rho_{Bj} = |U_{Bj}| - |V_B| + c(V_B, U_{Bj}). \quad (32)$$

Loops and parallel edges keep their usual multigraph meanings. In particular, each unobserved loop contributes one to the cycle rank.

Lemma 4.3 (Unit minors for cycle elimination). *A fundamental-cycle matrix with identity chord rows is totally unimodular. Let C be such a matrix, let D consist of d linearly independent rows, and choose d columns I such that $B = D_{:,I}$ is nonsingular. Write F for the complementary columns. Then, for every row c of C , the vectors*

$$W = c_I B^{-1}, \quad R = c_F - W D_{:,F} \quad (33)$$

have entries in $\{0, \pm 1\}$.

Proof. Delete one incidence row of the connected core and partition the reduced incidence matrix as $[N_T \ N_Q]$ into tree and chord columns. A tree basis has determinant ± 1 . Every square incidence minor is 0 or ± 1 : induction expands a column with at most one nonzero entry; if every column has two entries, they are opposite and the row sum is zero. Every minor of $N_T^{-1} N_Q$ equals, up to sign, the determinant of a matrix obtained by replacing the corresponding tree columns by chord columns, divided by $\det N_T$. This follows by multiplying the replacement matrix by N_T^{-1} and expanding its unchanged identity columns. The replacement is a square incidence submatrix. Therefore $-N_T^{-1} N_Q$ is totally unimodular. Appending identity chord rows preserves this property, proving the assertion for C . A core consisting of one loop has $C = (1)$ directly.

In particular $\det B = \pm 1$. Cramer's rule applied to $WB = c_I$ expresses each entry of W as a selected-row replacement minor of C divided by $\det B$. For a free column f , the bordered determinant identity gives

$$R_f = \frac{\det \begin{pmatrix} B & D_{:,f} \\ c_I & c_f \end{pmatrix}}{\det B}.$$

The numerator is a minor of C , or zero when its rows repeat. These formulas prove (33). Empty pivot or free sets have their usual zero-dimensional matrix interpretation. $\square$

Theorem 4.4 (Observation-sensitive extended hull). *There is an exact rational extended formulation for H with*

$$\sum_{B \in \mathcal{B}} \sum_{j \in J_B} \rho_{Bj} \quad (34)$$

additional variables and the row bound (28). Coefficients of x , z , and additional variables belong to $\{0, \pm 1\}$; simplex coefficients and right-hand sides have polynomial rational encoding length. The variable count is an upper bound for this construction, not a lower bound for arbitrary extended formulations.

Proof. Fix B and $j \in J_B$, suppress their subscripts, and write $C = \widehat{C}_B$, $r = r_B$. Choose a maximal independent set of observed path rows of C , giving $D \in \mathbb{R}^{d \times r}$. For each selected row choose one actual observed arc and use its value from (26), giving $q \in \mathbb{R}^d$. The selected entries of q involve distinct original product variables. All other observations will remain as equations, including any repeated path values.

Choose pivot columns I and complementary columns F as in lemma 4.3. Retain $g = t_F \in \mathbb{R}^{r-d}$ and eliminate the pivot coordinates by the reversible substitution

$$t_I = B^{-1}(q - D_{:,F}g). \quad (35)$$

For every core path p with row c_p define

$$\phi_{pj} = (c_{p,I}B^{-1})q + (c_{p,F} - c_{p,I}B^{-1}D_{:,F})g. \quad (36)$$

Replace $(\widehat{C}_B t^{B,j})_p$ by ϕ_{pj} in (29)–(31). Selected observation equations become identities and can be omitted; all other observation equations remain. Formula (35) proves equivalence in both directions, so no projection condition is lost.

By lemma 4.3, each selected product and retained variable has coefficient 0, 1, or -1 in every ϕ_{pj} . A nonselected observation adds a distinct product coordinate; it cannot add a second occurrence of a selected product. Different labels use disjoint product and auxiliary columns. The residual uses one original flow coordinate through $s_p(x)$. Thus collecting terms does not increase any x , z , or auxiliary coefficient beyond one in magnitude. Only coefficients of y collect reference-flow and interval data. Their encoding lengths remain polynomial. Elimination adds no rows.

It remains to identify $r - d$. Restrict a block circulation to O_{Bj} . The kernel is precisely the space of circulations supported on U_{Bj} , whose dimension is (32) by the incidence-rank formula. The core matrix parametrizes all block circulations, and observing any arc of a suppressed path fixes its signed path value. Therefore the rank of the restriction map is exactly the rank d of the observed core rows. Rank-nullity gives $r_B - d = \rho_{Bj}$, proving (34). $\square$

For clarity, the final block inequalities are

$$y_j L_p \leq \phi_{pj} \leq y_j U_p, \quad j \in J_B, p \in \mathcal{P}_B, \quad (37)$$

$$q_{ej} = \phi_{pj}, \quad (e, j) \in \mathcal{O}, e \in p \subseteq E_B, \quad (38)$$

$$\lambda_{B,*} L_p \leq s_p(x) - \sum_{j \in J_B} \phi_{pj} \leq \lambda_{B,*} U_p, \quad p \in \mathcal{P}_B. \quad (39)$$

The original flow and simplex domains and bridge observations remain part of the formulation. In particular, reconstructing state products by balance equations alone is insufficient: all displayed state and residual bounds must also be imposed.

The row counts do not bound matrix nonzeros when ranks grow. If $N = |V| + |E| + m + |\mathcal{O}|$ and $\mathcal{O}_B = \mathcal{O} \cap (E_B \times [m])$, both constructions have the explicit nonzero bound

$$O \left(N + \sum_{B: a_B > 0} [r_B^2(a_B + 1) + r_B |\mathcal{O}_B|] \right). \quad (40)$$

Indeed, a reconstructed path has $O(r_B)$ selected-product and auxiliary terms per label; there are $O(r_B)$ paths and a_B explicit labels. Each remaining observation also uses $O(r_B)$ terms. Hence bounded maximum block rank gives size linear in the sparse input, even as m and the graph grow. The whole formulation need not be totally unimodular. The unit coefficient statements use the displayed rational row scaling; clearing denominators in the simplex coefficients can destroy that unit bound in a primitive integer normalization.

4.4 Forest complements and product completion

Corollary 4.5 (An original-coordinate hull on arbitrary graphs). *Suppose (V_B, U_{Bj}) is a forest for every B and $j \in J_B$. Then (37)–(39), the original domains, and the bridge equations describe H with no additional variables. They have unit flow and product coefficients and the row bound (28).*

Proof. An undirected multigraph is a forest exactly when its cycle rank is zero. Thus every ρ_{Bj} in (34) vanishes. Apply theorem 4.4. $\square$

Equivalently, the observations for each locally observed label meet every cycle of its block. Entirely unobserved labels impose no condition: they are already included in the merged state. The condition is sufficient; it does not assert that remaining unobserved cycles prevent an original-coordinate description with unit coefficients.

Proposition 4.6 (Minimum individual-coordinate completion). *Fix B and $j \in J_B$. Among additions of individual missing arc-product coordinates, exactly ρ_{Bj} are necessary and sufficient to determine the full state circulation from the observed coordinates and linear balance equations on the ambient circulation space. A minimum choice consists of the nonforest edges of any spanning forest of (V_B, U_{Bj}) . The additional variables in theorem 4.4 may instead be chosen to represent these individual missing products, with the same variable and row bounds and unit flow, product, and auxiliary coefficients.*

Proof. The existing restriction map has nullity ρ_{Bj} . Each additional scalar coordinate can raise its rank by at most one, proving necessity. Let F_{Bj} be a spanning forest of the unobserved subgraph, including all its vertices. Add observations on $U_{Bj} \setminus F_{Bj}$, which has ρ_{Bj} edges. The remaining unobserved graph is a forest, so the restriction kernel is zero by the proof of theorem 4.4. This proves sufficiency and the stated minimum.

Apply corollary 4.5 to the completed observation set and regard the added product coordinates as auxiliary variables. Projection onto the original observations commutes with convexification, so this is an extended formulation for the original H . Completing labels only in J_B does not introduce new labels. The added equations number $\sum_{B,j} \rho_{Bj} \leq \sum_B r_B a_B$, already absorbed by the row bound. The corollary's coefficient property includes the added coordinates. $\square$

There is a direct interpretation of this completion. The block state flows satisfy $A_B f^{B,j} = y_j A_B v_{E_B}$, since their deviations are block circulations. Once the nonforest state products are known, sum these equations on either side of a cut formed by deleting a remaining forest edge from its component. That edge is the only unknown crossing edge, so its state product is an affine expression

in the known state products and the state weight, with unit product coefficients. Component balance equations enforce consistency if the unobserved forest is disconnected. This is network elimination of the kind underlying reduced RLT: Liberti and Pantelides [14, Theorem 3.1 and Section 2] reconstruct common-factor products by multiplying full-row-rank linear equations and retaining products on suitable complementary coordinates. Here the extra information is the observation-complement graph criterion and its exact sparse simplex-hull consequence.

The minimum in [proposition 4.6](#) concerns reconstruction by *individual coordinates on the ambient linear space*. It is neither an extension-complexity lower bound nor a claim about the number of coordinates needed on a capacity-degenerate slice. In particular, $\lambda_j = 0$ forces every state product to zero. The preprocessing in [lemma 4.1](#) can remove fixed arcs before the parameters are computed, but it does not remove this conditional-slice distinction.

All formulations also give explicit recovery. From retained coordinates, recover each $t^{B,j}$ using (35), reconstruct path deviations by $\hat{C}_B t^{B,j}$, and subtract their sum from $s(x)$ to recover the merged state. Expand deviations along the original paths using their signs. The proportional refinement (19) then supplies the global state flows and the at-most- $m + 1$ -point decomposition from [proposition 2.1](#). This recovery is exact rational arithmetic for rational input and a rational feasible extension; merely solving the formulation numerically does not provide an exact certificate.

Example 4.7 (Three observations on a K_4 block). Orient every edge of K_4 from the smaller to the larger vertex and take unit capacities and unit flow from vertex 1 to vertex 4. Thus $b = (-1, 0, 0, 1)^\top$. Use the reference flow $v_{14} = 1$ and zero on the other arcs. For one label j , observe 13, 14, and 24; the unobserved edges 12, 23, and 34 form a spanning tree. Writing $q_{13} = z_{13,j}$, $q_{14} = z_{14,j} - y_j$, and $q_{24} = z_{24,j}$, balance gives the remaining state deviations

$$h_{12} = -q_{13} - q_{14}, \quad h_{23} = -q_{13} - q_{14} - q_{24}, \quad h_{34} = -q_{14} - q_{24}.$$

For example, nonnegativity of the reconstructed flow on 23 gives

$$z_{13,j} + z_{14,j} + z_{24,j} \leq y_j. \tag{41}$$

This cut is absent from independent product convexification. To see the gap, take $m = 1$, $y_1 = 1/2$, $z_{13,1} = z_{14,1} = z_{24,1} = 1/5$, and let x average the four paths 1234, 124, 134, and 14 equally. Then $x_{12} = x_{34} = 1/2$ and the other four arc flows are $1/4$. All three observed products satisfy (5), but the left side of (41) is $3/5 > 1/2$. The complete hull requires all reconstructed bounds, including the residual bounds, not only this one cut.

5 Explicit hulls and separation on structured blocks

The extended formulations in [section 4](#) apply to arbitrary graphs. On some block classes, their state coordinates can be projected out explicitly while preserving unit flow and product coefficients. We first give a linear-arithmetic oracle for cycles and theta blocks, then a transportation oracle for any number of parallel paths. All graph classes refer to the underlying multigraph: orientations remain arbitrary, and different blocks may meet at articulation vertices. Bridges retain (15); independent loops use the scalar cycle formulas.

5.1 Cycle and theta state domains

A theta block consists of three internally vertex-disjoint paths between two terminals. Orient their traversals from the first terminal to the second. The three circulation deviations sum to zero, so

write them as $s, t, -s - t$. Put $h(s, t) = s + t$ and absorb the sign of the third path into σ_e : on the first two paths $\sigma_e = \varepsilon_e$, and on the third $\sigma_e = -\varepsilon_e$. Thus

$$x_e = v_e + \sigma_e \eta_e(s, t), \quad \eta_e \in \{s, t, h\}, \quad h = s + t. \quad (42)$$

The symbol h here is a linear coordinate form, not the number of states. Signed interval intersection as in (23) gives

$$T_B = \{(s, t) : L_s \leq s \leq U_s, L_t \leq t \leq U_t, L_h \leq s + t \leq U_h\}. \quad (43)$$

A cycle has the analogous scalar form $x_e = v_e + \sigma_e s$ and base interval $[L_s, U_s]$. Read the aggregate coordinates $s(x), t(x)$ from one representative arc on each of the first two paths; then $h(x) = s(x) + t(x)$. These representative arcs are distinct.

Fix a block and use its local states I_B and weights from (14). Suppress B in the local formulas below, so $\lambda_j = \lambda_{B,j}$, including the merged state $*$. Define $\tilde{q}_{ej} = \sigma_e(z_{ej} - v_e y_j)$ for observations in this block. Initialize and tighten each coordinate interval by

$$\ell_{\eta j} = \max(\{\lambda_j L_\eta\} \cup \{\tilde{q}_{ej} : (e, j) \in \mathcal{O}, \eta_e = \eta\}), \quad u_{\eta j} = \min(\{\lambda_j U_\eta\} \cup \{\tilde{q}_{ej} : (e, j) \in \mathcal{O}, \eta_e = \eta\}). \quad (44)$$

For $*$ there are no observations. Multiple observations on a path must agree: they all enter both endpoint lists. At zero weight the initial interval is the singleton zero, so feasibility forces every observed coordinate and every state deviation to be zero.

The cycle state domain is $[\ell_{sj}, u_{sj}]$. The theta state domain is

$$Q_j = \{(s, t) : \ell_{sj} \leq s \leq u_{sj}, \ell_{tj} \leq t \leq u_{tj}, \ell_{hj} \leq s + t \leq u_{hj}\}. \quad (45)$$

By theorem 3.2, block membership asks whether the aggregate coordinate belongs to the Minkowski sum of these local state domains. This reduction holds at every simplex boundary and for lower-dimensional state domains.

Lemma 5.1 (Theta feasibility and tight supports). *The state domain Q_j is nonempty if and only if*

$$\ell_{\eta j} \leq u_{\eta j} \quad (\eta = s, t, h), \quad \ell_{sj} + \ell_{tj} \leq u_{hj}, \quad \ell_{hj} \leq u_{sj} + u_{tj}. \quad (46)$$

When it is nonempty, its attained minimum and maximum of each coordinate form are

$$\begin{aligned} a_{sj} &= \max\{\ell_{sj}, \ell_{hj} - u_{tj}\}, & b_{sj} &= \min\{u_{sj}, u_{hj} - \ell_{tj}\}, \\ a_{tj} &= \max\{\ell_{tj}, \ell_{hj} - u_{sj}\}, & b_{tj} &= \min\{u_{tj}, u_{hj} - \ell_{sj}\}, \\ a_{hj} &= \max\{\ell_{hj}, \ell_{sj} + \ell_{tj}\}, & b_{hj} &= \min\{u_{hj}, u_{sj} + u_{tj}\}. \end{aligned} \quad (47)$$

Proof. The rectangle of individual coordinate intervals is nonempty exactly when its two intervals are nonempty. Its attainable sums fill $[\ell_{sj} + \ell_{tj}, u_{sj} + u_{tj}]$. Requiring that interval to intersect the nonempty sum interval gives precisely (46). For a fixed s , a feasible t exists if and only if $s \in [\ell_{sj}, u_{sj}]$ and $s \in [\ell_{hj} - u_{tj}, u_{hj} - \ell_{tj}]$. This proves both tight s bounds and their attainment. Interchanging s, t proves the next pair, and intersecting the two attainable sum intervals proves the last. $\square$

Lemma 5.2 (Six supports suffice for planar sums). *For any finite family of nonempty domains of the form (45), including segments and points,*

$$\sum_j Q_j = \left\{ (s, t) : \sum_j a_{\eta j} \leq \eta(s, t) \leq \sum_j b_{\eta j} \quad (\eta = s, t, h) \right\}. \quad (48)$$

The empty sum is the singleton zero, represented by six zero supports.

Proof. Let $\mathcal{N} = \{\pm(1, 0), \pm(0, 1), \pm(1, 1)\}$. Every edge of a two-dimensional Q_j has a normal in $\mathcal{N}$, because Q_j is defined by inequalities with those normals. A one-dimensional Q_j also has a direction perpendicular to a member of $\mathcal{N}$: at a relative interior point of the segment, at least one defining inequality is tight and its boundary contains the segment. Otherwise that point would have a two-dimensional neighborhood in Q_j .

For a linear objective with normal n , its maximizing face over a sum is the sum of its maximizing faces over the summands. Indeed, objective values add, and equality in the sum of the separate maximum values holds exactly when every summand attains its maximum. If the sum is two-dimensional, each exposed edge is therefore a sum of points and parallel segments, with at least one nontrivial segment. Its direction is a direction already occurring in a summand. Thus every edge normal of the sum lies in $\mathcal{N}$, and its support values are the sums of those in [lemma 5.1](#). These supports describe the polygon.

If the sum is a nontrivial segment, each nonpoint summand is a segment parallel to it. Its direction is one of the three permitted directions. The two normals perpendicular to that direction enforce its supporting line, and at least one of the other coordinate forms varies along that line. The lower and upper supports of that form give exactly the segment endpoints. If the sum is a point, the supports of s and t fix it. This proves the same six-inequality description in all dimensions. $\square$

The planar restriction matters. In higher dimension, an exposed facet can be a sum of lower-dimensional faces with different directions, so the preceding edge argument does not give a fixed list of facet normals. The support and normal-fan identities used here are classical [\[10\]](#); the explicit formulas exploit these particular block domains.

Theorem 5.3 (Cycle/theta original-coordinate hull). *Suppose every cyclic block is a cycle or a theta block. The hull H is described by the original flow and simplex constraints, the bridge equations, the cycle conditions*

$$\ell_{sj} \leq u_{sj} \quad (j \in I_B), \quad \sum_{j \in I_B} \ell_{sj} \leq s(x) \leq \sum_{j \in I_B} u_{sj}, \quad (49)$$

and, for every theta block, the local conditions [\(46\)](#) and six aggregate conditions

$$\sum_{j \in I_B} a_{\eta j} \leq \eta(s(x), t(x)) \leq \sum_{j \in I_B} b_{\eta j} \quad (\eta = s, t, h). \quad (50)$$

All endpoints are evaluated by [\(44\)](#). These expressions define a finite family of linear inequalities with coefficients in $\{0, \pm 1\}$ on x, z .

Proof. For intervals, nonemptiness is checked separately, and their Minkowski sum has the summed lower and upper endpoints. The theta case follows from [lemmas 5.1](#) and [5.2](#). The exact equivalence with the global hull then follows from [theorem 3.2](#).

Every lower endpoint in [\(44\)](#) is a maximum of affine functions, and every upper endpoint is a minimum. These properties also hold for the lower and upper supports in [\(47\)](#): for example, $\ell_{hj} - u_{tj}$ is a maximum of affine differences. Consequently each displayed condition can be written as a maximum of finitely many affine functions bounded above by zero. Requiring every affine branch to be at most zero is an equivalent finite linear description.

In a support branch, a state's product contribution uses either one coordinate type or two distinct types. Hence each observed product occurs at most once. Different states have different product indices. A local same-type interval comparison either uses two distinct observations or cancels an identical one; other local comparisons use distinct types. The aggregate $s(x), t(x)$ use separate representative arcs, and $h(x)$ uses their sum. Thus every flow or product coefficient is 0, 1, or -1 . Simplex coefficients contain the network data. The assertion concerns this rational row scaling, with the normalization qualifications given after [\(40\)](#). $\square$

5.2 A linear-arithmetic oracle and compact recovery

If any condition in [theorem 5.3](#) fails, select a maximizing affine branch of its convex piecewise-affine left side after moving all terms to one side. That branch is bounded above by zero at every hull point and has the same positive violation at the candidate. Thus endpoint and support comparisons produce a globally valid linear cut directly; they do not merely produce a local linear approximation. Ties may be resolved arbitrarily among active branches.

There are $a_B + 1$ local states in block B , and $\sum_B a_B \leq |\mathcal{O}|$. A constant number of endpoint and support operations suffices for each cycle/theta state after its observations have been scanned. Checking the original domains and all block conditions therefore uses

$$O(|V| + |E| + m + |\mathcal{O}|) \quad (51)$$

rational arithmetic operations, given observation indices grouped by block and label. Graph blocks, paths, and reference flows can be preprocessed in linear time. Sorting an ungrouped observation list adds $O(|\mathcal{O}| \log |\mathcal{O}|)$ comparisons if required. These are arithmetic and comparison counts, not constant-bit-cost claims. All evaluated rational expressions and emitted cut coefficients have polynomial encoding length.

Proposition 5.4 (Constructive cycle/theta recovery). *For a rational hull point in the scope of [theorem 5.3](#), a compact rational decomposition into at most $m+1$ graph points can be computed within (51), with the same grouping proviso. The compact representation stores the global weights, reference flow, one default coordinate per block, and exceptions for observed block/label pairs. Expanding all flow vectors can require $\Theta((m+1)|E|)$ output entries.*

Proof. Order the local states in a theta block and precompute suffix sums of the six tight supports. Suppose the remaining aggregate is $r = (r_s, r_t)$, with $r_h = r_s + r_t$, and the next state is i . Its permissible coordinate lies in $Q_i \cap (r - \sum_{j>i} Q_j)$. By [lemma 5.2](#), this is another theta polygon with intervals

$$\max \left\{ \ell_{\eta i}, r_{\eta} - \sum_{j>i} b_{\eta j} \right\} \leq \eta \leq \min \left\{ u_{\eta i}, r_{\eta} - \sum_{j>i} a_{\eta j} \right\}, \quad \eta = s, t, h. \quad (52)$$

It is nonempty because the remaining aggregate belongs to the remaining sum. In any nonempty theta polygon with interval endpoints ℓ, u , choose

$$s = \max\{\ell_s, \ell_h - u_t\}, \quad t = \max\{\ell_t, \ell_h - s\}. \quad (53)$$

By [lemma 5.1](#), the chosen s lies between its tight bounds, in particular $s \leq u_s$ and $s \leq u_h - \ell_t$. Both arguments defining t are at most u_t because $s \geq \ell_h - u_t$. The sum is at least ℓ_h ; if $t = \ell_t$ it is at most u_h , and otherwise it equals ℓ_h . Thus the chosen point is feasible. Subtract it from r and continue. Empty suffixes use zero supports. The scalar interval construction is the same intersection-and-subtraction procedure in one dimension.

Normalize each positive-weight explicit state coordinate by its weight. Normalize a positive-weight merged coordinate by its total weight and store it as the block's default for all constituent global states. If that weight is zero, the default is unused and may be the aggregate block coordinate, which is feasible because $x \in P$. An unobserved block also uses its aggregate coordinate as the default. The refinement proof of [theorem 3.2](#) combines these coordinates into feasible global flows with weights $\lambda_0, \dots, \lambda_m$; zero weights are omitted. The number of exceptions is at most $|\mathcal{O}|$, and all operations and stored coordinates obey the claimed bound. Enumerating a dense flow for every positive state is a separate output cost. $\square$

Example 5.5 (Separate state hulls miss a joint restriction). Consider three parallel arcs of unit capacity carrying unit total flow. Let their first two aggregate flows be $X_s = X_t = 1/3$, and let $y_1 = y_2 = 1/3$. Observe both first-two-arc products for both labels, and set all four to zero. For either label alone, the weighted state flow $(0, 0, 1/3)$ and its complementary flow $(1/3, 1/3, 0)$ give a feasible one-label hull decomposition. Thus the candidate belongs to the intersection of the two separately convexified one-label hulls.

Jointly, the two observed states would consume $2/3$ of the third arc, whose aggregate flow is only $1/3$. The violated joint inequality is

$$z_{s1} + z_{t1} + z_{s2} + z_{t2} \geq X_s + X_t - (1 - y_1 - y_2). \quad (54)$$

Its violation is $1/3$. It follows from the residual state's upper bound on the sum of the first two flows, and is included in (50). The example illustrates the known joint disaggregation mechanism rather than a separate convexification principle.

5.3 Any number of parallel paths

Now suppose each cyclic block, apart from independent loops, consists of $k \geq 2$ internally vertex-disjoint paths between two terminals. The number k is unrestricted. Orient all path traversals from the first terminal to the second, and use the signs ε_e of (22). The core domain is

$$L_i \leq \delta_i \leq U_i \quad (i \in [k]), \quad \sum_{i=1}^k \delta_i = 0. \quad (55)$$

Again write $\lambda_j = \lambda_{B,j}$ for local states, including $*$. For each local state $j \in I_B$, intersect its scaled path bounds with all observations on that path:

$$\ell_{ij} = \max(\{\lambda_j L_i\} \cup \{q_{ej} : (e, j) \in \mathcal{O}, e \in i\}), \quad u_{ij} = \min(\{\lambda_j U_i\} \cup \{q_{ej} : (e, j) \in \mathcal{O}, e \in i\}). \quad (56)$$

Here q_{ej} uses (26), without the theta-specific sign change. Let $\bar{\delta}_i = s_i(x)$ be the aggregate path deviations. Exact block membership is equivalent to finding a matrix $w = (w_{ij})$ such that

$$\ell_{ij} \leq w_{ij} \leq u_{ij}, \quad \sum_i w_{ij} = 0 \quad (j \in I_B), \quad \sum_j w_{ij} = \bar{\delta}_i \quad (i \in [k]). \quad (57)$$

This follows directly from [theorem 3.2](#).

Theorem 5.6 (Complete parallel-path subset description). *In this graph class, H is described by the original domains, bridge equations, scalar loop conditions, and the following conditions for every parallel-path block:*

$$\ell_{ij} \leq u_{ij} \quad (i, j), \quad \sum_i \ell_{ij} \leq 0 \leq \sum_i u_{ij} \quad (j \in I_B), \quad (58)$$

and, for every subset $S \subseteq [k]$,

$$\sum_{i \in S} \bar{\delta}_i \leq \sum_{j \in I_B} \min \left\{ \sum_{i \in S} u_{ij}, - \sum_{i \notin S} \ell_{ij} \right\}. \quad (59)$$

After expanding affine branches, this is a finite linear description whose coefficients on x, z lie in $\{0, \pm 1\}$, with the same rational-scaling qualification as [theorem 5.3](#).

Proof. The local conditions are necessary for (57). In a zero-sum column, the sum on S is bounded by its upper endpoints on S and by the negative lower endpoints on its complement. Summing over columns proves (59).

For sufficiency, shift all entries by their lower endpoints and define

$$g_{ij} = w_{ij} - \ell_{ij}, \quad C_{ij} = u_{ij} - \ell_{ij}, \quad r_i = \bar{\delta}_i - \sum_j \ell_{ij}, \quad c_j = - \sum_i \ell_{ij}. \quad (60)$$

We seek a nonnegative matrix $g \leq C$ with row sums r and column sums c . The local conditions give $C_{ij} \geq 0$ and $c_j \geq 0$. For each i , apply (59) to $[k] \setminus \{i\}$. Since $\sum_i \bar{\delta}_i = 0$, this gives

$$-\bar{\delta}_i \leq \sum_j \min \left\{ \sum_{h \neq i} u_{hj}, -\ell_{ij} \right\} \leq - \sum_j \ell_{ij},$$

so $r_i \geq 0$ as well. Moreover $R := \sum_i r_i = \sum_j c_j$.

Build a directed network with a source, one node for each row, one node for each column, and a sink. Source-to-row capacities are r_i , row-to-column capacities are C_{ij} , and column-to-sink capacities are c_j . A flow of value R saturates all source and sink arcs and yields exactly the desired matrix. Fixing the source-side row set S , the minimum cut over the side choices of column nodes has capacity

$$\sum_{i \notin S} r_i + \sum_j \min \left\{ \sum_{i \in S} C_{ij}, c_j \right\}. \quad (61)$$

Indeed, a sink-side column pays its incoming capacities from S , while a source-side column pays c_j ; the choices are independent. This cut has capacity at least R if and only if

$$\sum_{i \in S} r_i \leq \sum_j \min \left\{ \sum_{i \in S} C_{ij}, c_j \right\}.$$

Adding $\sum_{i \in S, j} \ell_{ij}$ to both sides converts this inequality exactly into (59). Hence every cut has capacity at least R . The maximum-flow/minimum-cut theorem [8] gives the desired flow, including the case $R = 0$, when all targets vanish. Undo the shift to obtain (57), then refine local states and combine blocks by theorem 3.2.

The right side of (59) is a minimum of finitely many affine expressions: u_{ij} and $-\ell_{ij}$ have this property, and it is preserved by sums and minima. Branch expansion therefore gives an equivalent finite linear family. In a branch for one state, either one upper endpoint from each path in S is used, or one negated lower endpoint from each complementary path is used. Each observation belongs to only one path and one state, so its coefficient has magnitude at most one. The local conditions have the same property, with possible cancellation of a repeated observation in a same-interval comparison. The aggregate left side uses distinct representative arcs. This proves the coefficient assertion. $\square$

For $k = 3$, the six nonempty proper subsets are the supports in (47), after setting $(\delta_1, \delta_2, \delta_3) = (s, t, -h)$. The empty and full subsets are redundant under (58). For larger k , the subset family may be exponential. Its completeness comes from the transportation projection, not from an extension of the planar edge argument. This is an application of classical transportation feasibility [9]; the network-Cayley projection of Kis and Horváth [13, Section 5.7] is a direct predecessor.

5.4 Separation by flow and reconstruction

The theorem supplies an oracle without enumerating the subsets. Check the original domains, bridge equations, and local conditions first. If a shifted row target r_i in (60) is negative, the valid condition

$$\bar{\delta}_i \geq \sum_j \ell_{ij} \quad (62)$$

separates the point; its validity was proved from the complementary subset above. Otherwise all transportation capacities and targets are nonnegative. Compute a maximum flow. Value R proves block membership and returns its state matrix. A smaller value gives a minimum cut; its source-side row set violates (59), because minimizing column sides can only decrease the cut capacity.

To return an original-coordinate linear inequality, select the active minimum alternatives and active affine endpoint expressions in the violated subset condition. Their sum is an affine majorant of its concave right side and agrees with it at the candidate. Hence the resulting affine upper bound is globally valid and strictly violated. Local violations and (62) are handled by their active affine branches in the same way.

For a block with k_B paths and a_B observed labels, the transportation network has

$$k_B + a_B + 3 \text{ nodes}, \quad k_B(a_B + 1) + k_B + a_B + 1 \text{ arcs.} \quad (63)$$

Graph and endpoint preprocessing, target construction, and cut extraction use

$$O\left(|V| + |E| + m + |\mathcal{O}| + \sum_B k_B(a_B + 1)\right) \quad (64)$$

arithmetic operations after observation grouping, followed by one maximum flow computation per parallel-path block. A polynomial-time algorithm such as shortest augmenting paths [7] gives polynomial bit complexity for rational data. One way to see the encoding bound is to clear the finitely many capacity denominators with a common multiple; its encoding length is at most their summed encoding lengths. Flow updates then use integers bounded by the total capacity. An arbitrary augmenting-path rule is not needed for this complexity claim. When k_B grows, (64) and the flow solves replace the linear cycle/theta bound.

The returned matrices provide exact rational block state deviations. Normalize positive-weight columns and store the normalized merged column as the default for its constituent global states. Zero-weight columns vanish by their original scaled path bounds, and no division by zero occurs. Together with the reference flow and block paths, the resulting compact decomposition has storage of order (64); expanding every dense state-flow vector can still require $\Theta((m + 1)|E|)$ entries.

The cycle rank of a parallel-path block is $k_B - 1$, so this theorem permits unbounded rank within a block. It does not cover every series-parallel block. In particular, several serial parallel gadgets inside one outer bypass can have interacting internal state profiles that are not represented by the single transportation matrix (57). The present results supply explicit structural oracles and unit-coefficient descriptions; generic polynomial separation already follows from [proposition 2.1](#).

6 Separation and coefficient bounds at bounded block rank

The preceding unit-coefficient oracles rely on special block shapes. We now allow arbitrary blocks and bound their cycle ranks instead. The construction uses finite libraries determined by each block's signed cycle normals. It has no online linear-programming subproblem. Its parameter

dependence is large, so it provides a structural bound rather than a runtime advantage over the compressed formulations.

For $s \geq 1$ define

$$H_1 = 1, \quad H_s = \max\{1, \lfloor (s-1)^{(s-1)/2} \rfloor\} \quad (s \geq 2). \quad (65)$$

The first four values are 1, 1, 2, 5. Put $r = \max_{B \in \mathcal{B}} r_B$. If there are no cyclic blocks, the flow and simplex domains and bridge equations already describe the hull, so no rank-dependent procedure is needed.

6.1 A fixed normal matrix for every state

Fix a cyclic block of rank s . Use original chord arcs as cycle coordinates, with the fundamental-cycle matrix $C \in \{0, \pm 1\}^{E_B \times s}$ of [lemma 3.1](#), so $x_{E_B} = v_{E_B} + C\theta$ and the chord rows are the identity. Let $\mathcal{A}$ contain the distinct nonzero rows of C with both signs, and let M be the matrix with row set $\mathcal{A}$. It is totally unimodular by [lemma 4.3](#), has rank s , contains $\pm e_i$, and has at most $3^s - 1$ rows. Repeated normals are combined by defining

$$\beta_a = \min(\{u_e - v_e : C_e = a\} \cup \{v_e : -C_e = a\}), \quad T = \{\theta : M\theta \leq \beta\}. \quad (66)$$

The lists in this formula are nonempty by construction. Every arc in a cyclic block has a nonzero cycle row, since it lies on a cycle. Hence these inequalities include all arc bounds. The domain T is bounded and nonempty when P is nonempty.

For each local state $j \in I_B$, write $\lambda_j = \lambda_{B,j}$ and set

$$\gamma_{aj} = \min(\{\lambda_j \beta_a\} \cup \{z_{ej} - y_j v_e : C_e = a, (e, j) \in \mathcal{O}\} \cup \{-z_{ej} + y_j v_e : -C_e = a, (e, j) \in \mathcal{O}\}). \quad (67)$$

The observation lists are empty for the merged state. Each γ_{aj} is a concave piecewise-affine function of the original coordinates. Write

$$Q_j = \{\theta : M\theta \leq \gamma_j\}, \quad \bar{\theta}_i = x_{e_i} - v_{e_i}, \quad (68)$$

where e_i is chord i . With the original domains and bridge equations enforced, [theorem 3.2](#) reduces hull membership to

$$Q_j \neq \emptyset \quad (j \in I_B), \quad \bar{\theta} \in \sum_{j \in I_B} Q_j \quad (69)$$

for each block. At zero weight, the signed coordinate bounds in [\(67\)](#) force $\theta = 0$, or reveal an inconsistent observation. Thus all subsequent statements include zero weights.

6.2 Finite local feasibility certificates

Let $\mathcal{K}$ contain the primitive generators of the extreme rays of

$$K = \{q \in \mathbb{R}_+^{|\mathcal{A}|} : M^\top q = 0\}. \quad (70)$$

These are the positive circuits of the row configuration.

Lemma 6.1 (Unit local certificates). *Every $q \in \mathcal{K}$ is the indicator vector of a minimally dependent row set of size at most $s + 1$. Moreover,*

$$Q_j \neq \emptyset \iff q^\top \gamma_j \geq 0 \quad (q \in \mathcal{K}). \quad (71)$$

A violated test yields an affine original-coordinate cut with unit product coefficients by selecting active branches in [\(67\)](#).

Proof. For an extreme nonnegative dependence, the dependence space on its support must be one-dimensional. Otherwise a linearly independent dependence could perturb its strictly positive support in both directions while remaining nonnegative, contradicting extremality. No proper subset is dependent: such a dependence would provide the same perturbation unless it were a multiple of the original vector, which has full support. Hence the support has at most $s + 1$ rows. A cofactor formula on a full-rank set of columns of this minimally dependent row matrix expresses its primitive dependence by nonzero minors. Total unimodularity makes every nonzero coefficient ± 1 , and positivity makes them all 1.

Farkas' lemma says $M\theta \leq \gamma_j$ is feasible exactly when $q^\top \gamma_j \geq 0$ for every $q \in K$. The pointed polyhedral cone K is generated by its extreme rays, proving (71). Enumerating row subsets of size at most $s + 1$ and retaining the one-dimensional positive minimal dependencies therefore constructs the entire finite library.

An observation occurs only in the two opposite row normals C_e and $-C_e$. If a minimal positive dependence uses both these normals, it is exactly that opposite pair. Selecting the same observation twice then cancels it. Otherwise each observation occurs at most once. Selecting active affine minima gives an affine upper bound on $q^\top \gamma_j$ that equals its negative value at the candidate and is nonnegative on the hull. Its product coefficients are thus $0, \pm 1$. $\square$

6.3 Edge directions and a complete support library

For each set of $s - 1$ independent rows of M , take one orientation of its primitive integer null vector and deduplicate. Call the resulting set $\mathcal{D}$. For $s = 1$ set $\mathcal{D} = \{1\}$.

Lemma 6.2 (Uniform edge directions). *Every direction $d \in \mathcal{D}$ satisfies*

$$d \in \{0, \pm 1\}^s \setminus \{0\}, \quad Md \in \{0, \pm 1\}^{|\mathcal{A}|}. \quad (72)$$

The set has at most $(3^s - 1)/2$ elements and includes each coordinate direction. It contains every edge direction of every nonempty Q_j , including lower-dimensional states.

Proof. The cofactor null vector of $s - 1$ independent rows has entries $0, \pm 1$ by total unimodularity, and at least one entry is nonzero. It is therefore already primitive. Its scalar product with another row is, up to sign, the full determinant obtained by adjoining that row, again $0, \pm 1$. This proves (72) and the cardinality bound. Using all but one coordinate row gives each coordinate direction.

At a relative interior point of a one-dimensional face, the tight defining rows have rank $s - 1$. To justify this even without interior, their common nullspace is exactly the direction space of the minimal face: all nontight inequalities remain slack under sufficiently small motions in that nullspace. Rows tight throughout Q_j are included among them. Choosing $s - 1$ independent tight rows gives the asserted edge direction. Point states have no edges to check. $\square$

Form the central arrangement in objective space with hyperplanes $d^\top h = 0$, $d \in \mathcal{D}$. Let $\mathcal{R}$ contain both primitive orientations of the null vector of every $s - 1$ independent directions from $\mathcal{D}$; for $s = 1$ use $\mathcal{R} = \{-1, 1\}$. Extra directions in this list cause no difficulty. The coordinate hyperplanes occur, so every full-dimensional closed chamber is pointed. Every extreme ray of such a chamber is generated by a member of $\mathcal{R}$, since its tight hyperplanes have rank $s - 1$. Cofactors and Hadamard's determinant bound give

$$|h_i| \leq H_s \quad (h \in \mathcal{R}). \quad (73)$$

Indeed each cofactor has order $s - 1$ and ternary entries; dividing by the greatest common divisor to make it primitive cannot increase it.

Lemma 6.3 (Finite support characterization). *For any finite family of nonempty bounded polytopes whose edge directions belong to $\mathcal{D}$, membership in their sum is equivalent to*

$$h^\top \bar{\theta} \leq \sum_j \sigma_j(h) \quad (h \in \mathcal{R}), \quad \sigma_j(h) = \max_{\theta \in Q_j} h^\top \theta. \quad (74)$$

This includes lower-dimensional summands and sums. The empty sum has support zero on every ray and is the singleton zero.

Proof. In the interior of a chamber, an objective is perpendicular to no possible edge. Its maximizing face in each summand is therefore a single vertex: a positive-dimensional face of a polytope has an edge. Along a line segment inside the chamber, the maximizing vertex cannot change without a nonunique maximizer at an intermediate objective. Thus each support function is linear on the chamber, and by continuity on its closure. Its sum is linear there as well. Every vector in a pointed polyhedral chamber is a nonnegative combination of its extreme-ray generators. The inequalities on $\mathcal{R}$ therefore imply the support inequality for every objective in every chamber closure, which covers all objective space. The support characterization of a nonempty compact convex set completes the proof. For the empty sum the same argument gives all inequalities $h^\top \bar{\theta} \leq 0$, forcing zero. $\square$

This is the classical edge-direction/zonotope refinement argument [10, 16, 17]. Network-flow edge directions are also studied by Onn et al. [18]. The proof is included because uniformity over the state right-hand sides and lower-dimensional slices is essential here.

6.4 Exact support formulas and bounded dual multipliers

For $h \in \mathcal{R}$, let $\mathcal{B}(h)$ contain every nonsingular s -row submatrix B of M for which

$$q_B = B^{-\top} h \geq 0. \quad (75)$$

Rows and components have the same chosen order; extending q_B by zero gives a multiplier on all rows of M .

Lemma 6.4 (Support bases and multiplier bound). *For every nonempty state domain,*

$$\sigma_j(h) = \min_{B \in \mathcal{B}(h)} q_B^\top \gamma_{B,j}. \quad (76)$$

The list $\mathcal{B}(h)$ is nonempty, and every retained multiplier is an integer satisfying

$$0 \leq (q_B)_i \leq H_s. \quad (77)$$

Proof. The support LP has dual $\min\{q^\top \gamma_j : M^\top q = h, q \geq 0\}$. Signed coordinate rows make it feasible for every h . The primal is nonempty and bounded, so duality gives a finite attained optimum. There is an optimal solution with linearly independent supporting columns of $M^\top$. Indeed, a dependence on a positive support permits both signs of a small feasible perturbation; at an optimum its objective change must be zero. Moving until a component vanishes reduces the support, and repetition yields independence. Extend those rows to a basis of $\mathbb{R}^s$; the added multipliers are zero. This proves (76), also for a degenerate or lower-dimensional primal.

Total unimodularity gives $\det B = \pm 1$, so q_B is integral. The stronger magnitude bound uses the defining directions of the ray rather than a triangle inequality on $B^{-\top} h$. Choose independent $d_1, \dots, d_{s-1} \in \mathcal{D}$ whose primitive orthogonal vector is h . By (72), every Bd_i is ternary, and these vectors remain independent. The integer vector $B^{-\top} h$ is orthogonal to them and primitive, because a unimodular transformation preserves primitive integer vectors. Their cofactor vector, divided by its common divisor, therefore equals $B^{-\top} h$ up to sign. Hadamard's bound proves (77). Rank one is immediate. $\square$

Theorem 6.5 (Bounded-rank separation and coefficients). *For maximum cyclic block rank r , the sparse hull has a finite linear description whose flow and observed-product coefficients are integers of magnitude at most H_r . The sharper bound H_{r_B} applies to inequalities from block B . Simplex coefficients may contain rational network data; the bound is in the specified rational row scaling.*

After observation grouping, exact membership and original-coordinate linear separation require

$$2^{O(r^2)}(|V| + |E| + m + |\mathcal{O}|) \quad (78)$$

rational arithmetic operations, including construction of finite block libraries. No online LP is required. Ungrouped observations may add $O(|\mathcal{O}| \log |\mathcal{O}|)$ sorting comparisons.

Proof. Use the original domains and bridge equations, the local circuit tests (71), and the support tests (74), evaluating each support by (76). Their equivalence to (69) proves completeness. Each support expression is a minimum of affine functions, because its multipliers are nonnegative and its endpoints are affine minima. If a support test fails, choose a minimizing basis and active endpoint branches separately for each state. The resulting affine majorant agrees with the support sum at the candidate, so it is a valid violated cut.

A nonsingular row basis cannot contain opposite normals. An observation therefore appears at most once in a state's selected support expression, with coefficient magnitude at most H_s by lemma 6.4. Product indices differ between states. The aggregate uses distinct original chord coordinates with coefficients h_i , bounded by (73). Local circuit cuts have unit product coefficients, and the original domains and bridge equations have unit flow/product coefficients. Expanding all branches gives the asserted finite description. Adding or clearing rational simplex denominators is not part of this normalization.

There are at most $3^s - 1$ signed normals and $(3^s - 1)/2$ edge directions. Enumerating their subsets of size at most $s + 1$, all candidate support rays, and all normal bases takes $2^{O(s^2)}$ operations and storage. The product of the ray and basis counts has the same bound. All library integers and their encoding lengths depend only on s .

Fundamental cycles can be constructed in $O(r|E|)$ graph/arithmetic operations. Repeated normals are combined in parameter-dependent work per arc, and each observation updates two endpoint entries. A block with a_B observed labels then needs $2^{O(r_B^2)}(a_B + 1)$ arithmetic operations to initialize and check its local states and evaluate all supports. Since $\sum_B a_B \leq |\mathcal{O}|$ and there are at most $|E|$ blocks, summing these bounds proves (78). $\square$

Online operations are rational affine evaluations, comparisons, and sums with parameter-bounded integer multipliers. Their encoding lengths are polynomial in the total input/candidate encoding length and the library encoding length. The arithmetic bound does not charge unbounded integers as single-bit objects. It also does not assert that a universal library is computationally attractive at moderate rank: the parallel-path oracle can be preferable even when its block ranks are unbounded.

6.5 Constructive recovery without an online LP

The finite supports can also provide a direct constructive decomposition. The following method deliberately uses a larger parameter factor than separation.

Theorem 6.6 (Finite-basis compact recovery). *A rational hull point with maximum cyclic block rank r admits a compact rational decomposition into at most $m + 1$ graph points, computable without an online LP in*

$$2^{O(r^3)}(|V| + |E| + m + |\mathcal{O}|) \quad (79)$$

arithmetic operations, including parameter-dependent preprocessing and with the same observation-grouping proviso as [theorem 6.5](#). The output and intermediate rational numbers have polynomial encoding length in the input/candidate and library encoding lengths. Dense state-flow output has the separate cost stated in [proposition 5.4](#).

Proof. Fix a block of rank s . Compute every state support on $\mathcal{R}$ as above, order the local states, and precompute suffix sums $S_i(h) = \sum_{j>i} \sigma_j(h)$. At step i , let t be the remaining aggregate, which by induction belongs to $\sum_{j \geq i} Q_j$. The permissible next state vector is in the nonempty polytope

$$W_i(t) = \{\theta : M\theta \leq \gamma_i, \quad -h^\top \theta \leq S_i(h) - h^\top t \quad (h \in \mathcal{R})\}. \quad (80)$$

By [lemma 6.3](#), this is exactly $Q_i \cap (t - \sum_{j>i} Q_j)$. It is bounded because it is contained in Q_i . Even if lower-dimensional, a nonempty bounded polytope has a vertex, and its active defining normals at a vertex have full ambient rank: otherwise a nonzero null direction would allow small motions in both directions without violating any inequality.

The normal list N of (80) consists only of the rows of M and $-\mathcal{R}$ and is fixed before the candidate is given. Enumerate all nonsingular s -row submatrices of N and precompute their inverses. For each basis, solve its equations using the current right-hand side and test the resulting vector against every row of (80). At least one candidate is a feasible vertex by the preceding paragraph. Choose the first, subtract it from t , and continue. The last suffix is the singleton zero, so the construction ends with zero remaining aggregate. This proves exact state recovery without solving an online LP.

The normal list has $2^{O(s^2)}$ rows. Its s -row basis enumeration, including all feasibility checks per step, costs $2^{O(s^3)}$ arithmetic operations and storage. The earlier support computation is cheaper. There are $a_B + 1$ local states per block, so summing the same sparse state-count bound as in [theorem 6.5](#) proves (79).

For completeness, sequential division does not hide an exponential encoding cost in the number of states. The fixed normal entries are integers bounded by H_s . Every inverse basis is its integer adjugate divided by a nonzero integer determinant of parameter-bounded encoding length. All initial state endpoints and supports have a common rational denominator whose encoding length is polynomial in the input/candidate length and the library length. At each chosen step, a common denominator for the remaining aggregate need only be multiplied by that step's basis determinant. After $a_B + 1$ steps its encoding length has therefore grown additively by at most $a_B + 1$ parameter-bounded terms. Numerators also have polynomial length: state points are within their finite base bounds, remaining aggregates are sums of such points, and rejected basis candidates are obtained from these right-hand sides by fixed parameter-bounded matrices. This proves the claimed intermediate and output encoding bound.

Finally, normalize positive-weight explicit state coordinates and use the normalized merged state as the default for its constituent global labels. The zero-weight and unused-default conventions of [proposition 5.4](#) apply unchanged. Expand through C and use [theorem 3.2](#) to obtain graph points with the global simplex weights. A block coordinate has s entries, so defaults and observed-label exceptions give a compact representation within the stated bound. $\square$

6.6 Coordinate sections and necessary coefficients

We next state a coefficient-transfer observation that will also apply to the more general network constructions below. A *coordinate plane* fixes all original coordinates except two designated coordinates U, V ; it does not rescale or project additional free coordinates.

Lemma 6.7 (Local section forces an ambient coefficient ratio). *Let P be a polytope and L a coordinate plane with free coordinates U, V . Suppose that in an open neighborhood of a point (U_0, V_0) , the section $P \cap L$ is exactly*

$$\alpha U + \beta V \leq \gamma, \quad \alpha U_0 + \beta V_0 = \gamma, \quad (\alpha, \beta) \neq (0, 0). \quad (81)$$

Every finite linear inequality description of P , together with valid affine equations, contains an inequality whose coefficients on U, V are a positive multiple of (α, β) . In particular, a relative facet description contains such an inequality. Adding any equation valid on P to a facet representative cannot change those two coefficients.

Proof. The section has two-dimensional interior near (U_0, V_0) . Consequently every valid affine equation restricts to an identity on L : its coefficients on U, V are zero, and its fixed-coordinate constant is zero. Restrict a finite inequality description to L . At the boundary point, at least one nonconstant restricted inequality must be active. Otherwise all nonconstant inequalities are strictly satisfied in a neighborhood, and constant inequalities cannot produce the asserted boundary. An active valid inequality must vanish on the local boundary line, because feasible points occur in both tangent directions. Its normal is therefore proportional to (α, β) , and feasibility on the interior side makes the multiple positive. The argument applies to a finite relative facet description as well. The previously proved zero coefficients of valid equations establish representative invariance. $\square$

6.7 Sharp rank-three examples with sparse observations

Use vertices 0, 1, 2, 3 with every edge of K_4 directed toward the larger vertex, order the arcs as (12, 13, 23, 01, 02, 03), and set every capacity to one. The tree is 01, 02, 03, with chord matrix

$$C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & -1 \end{pmatrix}. \quad (82)$$

The following construction uses only two explicit simplex variables and five observed products. Unlike the source–sink example in [example 4.7](#), it has multiple supplies and demands.

Take reference flow and balances

$$v = (1/2, 1/2, 1/2, 1/4, 1/2, 1/2), \quad b = Av = (-5/4, -3/4, 1/2, 3/2)^\top. \quad (83)$$

Fix $y_1 = y_2 = 1/3$, so the residual weight is also $1/3$, and fix

$$\bar{\theta} = (1/6, 1/24, 1/8), \quad \bar{x} = (2/3, 13/24, 5/8, 11/24, 11/24, 1/3). \quad (84)$$

Observe exactly

$$\mathcal{O} = \{(12, 1), (13, 1), (02, 1), (23, 2), (01, 2)\}. \quad (85)$$

Leave $U = z_{12,1}$ and $V = z_{13,1}$ free, and fix the other observations to $z_{02,1} = z_{23,2} = 1/6$ and $z_{01,2} = 1/12$. Put $p = U - 1/6$, $q = V - 1/6$. The observations give

$$\theta^1 = (p, q, p), \quad \theta^2 = s(1, -1, 0). \quad (86)$$

All state deviations on arcs other than 01 lie in $[-1/6, 1/6]$; the interval on 01 is $[-1/12, 1/4]$. In particular, the residual state obeys $\theta_1^0 \leq 1/6$ and $\theta_2^0 + \theta_3^0 \leq 1/6$. For $h = (1, 1, 1)$, these bounds give $h^\top \theta^0 \leq 1/3$, while $h^\top \theta^2 = 0$ and $h^\top \bar{\theta} = 1/3$. Every feasible point in the section therefore satisfies

$$2p + q \geq 0, \quad 2U + V \geq 1/2. \quad (87)$$

Conversely, suppose $|p|, |q| < 1/96$ and $w = 2p + q \geq 0$. Choose

$$s = q/2, \quad \theta^0 = (1/6 - p - q/2, 1/24 - q/2, 1/8 - p). \quad (88)$$

These vectors sum to (84). The first state's arc deviations are $(p, q, p, p + q, 0, -p - q)$, all within its scaled bounds. The second state's arc deviations have magnitudes at most $|q|/2$ and are zero on 01. The residual arc deviations are

$$C\theta^0 = (1/6 - w/2, 1/24 - q/2, 1/8 - p, 5/24 - p - q, -1/24 + q/2, -1/6 + w/2). \quad (89)$$

Here $0 \leq w < 1/32$. The first and last entries lie in their closed intervals $[-1/6, 1/6]$. The second, third, and fifth entries lie strictly inside those intervals. The fourth lies in $(3/16, 11/48)$, contained in $[-1/12, 1/4]$. Thus all scaled bounds hold, as do the balances and observations. Disaggregation proves that the section is exactly the half-plane (87) in the asserted neighborhood.

Corollary 6.8 (Rank-three sharpness with two simplex variables). *The bound $H_3 = 2$ in theorem 6.5 cannot be replaced by one, even for a unit-capacity acyclic K_4 with two explicit simplex variables and five observed products. Every finite original-coordinate description of this hull contains an inequality whose coefficients on U, V have ratio two.*

Proof. Apply lemma 6.7 with the local normal $(-2, -1)$ at $(U, V) = (1/6, 1/6)$. An integer flow/product coefficient description with all magnitudes at most one cannot contain this ratio, whereas theorem 6.5 supplies a description with bound two. $\square$

Remark 6.9 (A symmetric-reference variant). There is also a seven-observation construction with the same graph and C , the uniform reference $v = (1/2, \dots, 1/2)$, and balances $(-3/2, -1/2, 1/2, 3/2)^\top$. Use three explicit variables $y_1 = y_2 = y_3 = 1/4$, aggregate cycle vector $(1/8, 0, 1/8)$, and therefore aggregate flows $(5/8, 1/2, 5/8, 5/8, 1/2, 3/8)$. Observe

$$(12, 1), (13, 1), (02, 1), (12, 2), (03, 2), (23, 3), (01, 3),$$

fix all but $U = z_{12,1}, V = z_{13,1}$ to $1/8$, and put $p = U - 1/8, q = V - 1/8$. The explicit states are $(p, q, p), t(0, 1, -1)$, and $s(1, -1, 0)$; every state has $|C\theta^j| \leq 1/8$. The residual support in direction $(1, 1, 1)$ is at most $1/4$, so $2p + q \geq 0$, or $2U + V \geq 3/8$, is necessary. For $|p|, |q| < 1/32$ and $w = 2p + q \geq 0$, the choices

$$t = -q/2, \quad s = q/2, \quad \theta^0 = (1/8 - w/2, 0, 1/8 - w/2)$$

prove sufficiency locally. The first state's arc deviations are $(p, q, p, p + q, 0, -p - q)$, the two line-state magnitudes are less than $1/64$, and the residual arc deviations are $(a, 0, a, a, 0, -a)$ with $5/64 < a = 1/8 - w/2 \leq 1/8$. This variant has completely symmetric state bounds. The five-product construction changes v_{01} and the associated balances $b = Av$, hence changes the flow domain itself. This modification makes the residual support face an edge instead of a point, allowing one fewer explicit line state. Changing only the reference within one fixed flow domain would merely translate its cycle-coordinate representation.

This necessity result concerns actual retained product coordinates and is invariant under adding the hull's affine equations. It is stronger than finding a dual basis with multiplier two. The related example of Khademnia and Davarnia [12, Example 2] exhibits a projection-cone aggregation weight two on a different graph, with equality balances represented by pairs of opposite inequalities; it is an antecedent for weighted cancellation, but does not by itself give this equality-network facet-ratio conclusion.

7 Coordinate sections and unrestricted coefficient growth

The bounded-rank hypothesis cannot be removed from the coefficient bounds. We first transfer transportation universality into the original coordinates of a sparse bilinear hull. The underlying universality theorem is classical: De Loera and Onn [6, Theorem 1.1] represent every bounded rational standard-form polytope by a slim three-way transportation polytope, with a coordinate-erasing bijection. Their construction in Section 3.3, printed page 816, places the retained coordinates in the first layer: its injection is $(i, j, k) \mapsto ((i, j), (1, k), 1)$. Composing the coordinate injections in their construction therefore retains any specified original coordinates in that layer. Their Theorem 1.2 and discussion on pages 807–808 already establish bitransportation universality and its two-commodity network interpretation. The statement below concerns the further transfer to a *sparse bilinear coordinate section*, not universality of multicommodity networks itself.

Theorem 7.1 (Sparse coordinate-section transfer). *Let $Q \subseteq \mathbb{R}_+^p$ be a nonempty bounded rational polytope, given by a rational inequality description. In polynomial time one can construct an acyclic four-layer network, a sparse observation set with two explicit simplex variables, and a coordinate affine subspace fixing every original hull coordinate except p observed products, such that the hull section in these p coordinates is exactly Q . All arcs have the same positive integer capacity B , and the flow value is B . After dividing every flow and product coordinate by B , all capacities and the flow value are one and the section becomes Q/B .*

Proof. Introduce nonnegative slack variables for the inequalities of Q and clear equation denominators. Since Q is bounded, its standard-form extension is bounded as well. Apply the preceding transportation theorem and its first-layer coordinate injection. Write the resulting table polytope as

$$\mathcal{T} = \left\{ t \geq 0 : \sum_{k=1}^3 t_{ijk} = U_{ij}, \quad \sum_j t_{ijk} = V_{ik}, \quad \sum_i t_{ijk} = W_{jk} \right\}, \quad (90)$$

where $i \in [r]$, $j \in [c]$, and all margins are nonnegative integers. There are p distinct designated cells $(i, j, 1)$ whose values range exactly over Q ; erasing the other cells includes erasing the slack coordinates. Nonemptiness implies consistency of the margins.

We may require $D_k := \sum_i V_{ik} = \sum_j W_{jk} > 0$ for every layer. Indeed, append one row and one column, put all old–new cross-cell two-dimensional margins equal to zero, and put the new corner margin equal to three. Assign new row and column layer margins equal to one in each layer. Nonnegativity forces all cross entries to zero and each corner entry to one. This padding does not change any old cell and increases each D_k by one. In the following, r, c include the padding.

Put $B = D_1 + D_2 + D_3$ and build the network

$$s \longrightarrow r_i \longrightarrow c_j \longrightarrow t \quad (i \in [r], j \in [c]),$$

with every indicated arc present, every capacity B , and net supply B at s and demand B at t . Observe labels 1 and 2 on every boundary arc sr_i, c_jt . On interior arcs observe label 1 only at the

p designated cells. Thus there are $rc + r + c$ arcs and $2(r + c) + p$ observed products. Fix the coordinates

$$\begin{aligned} x_{r_i c_j} &= U_{ij}, & x_{sr_i} &= \sum_k V_{ik}, & x_{c_j t} &= \sum_k W_{jk}, \\ y_k &= D_k/B, & z_{sr_i, k} &= V_{ik}, & z_{c_j t, k} &= W_{jk} \quad (k = 1, 2). \end{aligned}$$

These fix every original coordinate other than the designated interior products. The residual weight is D_3/B .

By [proposition 2.1](#), a point of this section has three state flows. Their interior entries give a table satisfying (90): the aggregate interior flows give U , the two observed boundary layers give V, W for $k = 1, 2$, and subtraction from the aggregate gives the third layer. Conversely any table supplies the three state flows, including their boundary arcs. Every arc of a nonnegative flow of value D_k on this acyclic network has flow at most D_k ; for example, decompose it into directed source–sink paths by repeatedly subtracting a positive path flow. Hence the scaled capacity bounds $f_e^k \leq (D_k/B)B$ hold. The designated products are precisely the designated table entries. This proves exactness without first forming a hull that retains all interior products and then asserting that its facets survive projection.

All transformations have polynomial output size and rational encoding length. The padding adds only one row and one column. Finally the invertible change $x' = x/B$, $z' = z/B$, $y' = y$ converts the network to unit capacities and unit flow value, and scales the free section coordinates by $1/B$. $\square$

Corollary 7.2 (Large coefficients with two simplex variables). *For every integer $M \geq 2$ there is a sparse bilinear hull on an acyclic unit-capacity unit-flow network, with $m = 2$, whose every finite original-coordinate inequality description contains an inequality with two observed-product coefficients in ratio M . This ratio is unchanged by adding affine-hull equations. The instances can be chosen to have encoding length polynomial in $\log M$, with all numerical data in the original nonlinear model in $\{-1, 0, 1\}$. In particular, the magnitudes required in integer inequality descriptions are not polynomially bounded by the original input length.*

Proof. Apply [theorem 7.1](#) to $Q_M = \{(U, V) \geq 0 : U + MV \leq 1\}$. After normalization the free coordinate section is $\{(U, V) \geq 0 : U + MV \leq 1/B\}$. At an interior point of its sloping edge the section is locally exactly one half-plane. [Lemma 6.7](#) gives the ratio M , including invariance under affine-hull equations. Two nonzero integer coefficients in this ratio have maximum magnitude at least M . The transportation construction has polynomial size in the encoding length of Q_M , which is $O(\log M)$; normalization leaves only the incidence, unit capacities, unit balances, simplex constraints, and selected bilinear equations in the original model. The rational constants fixing the section are not additional data of that model. Taking $M = 2^k$ proves the last assertion. $\square$

These hulls are 0/1 polytopes. A nonnegative unit flow on an acyclic network is a convex combination of path incidence vectors. Pairing each path with each simplex vertex gives 0/1 values of x, y, z , and [proposition 2.1](#) shows that their convex hull is exactly the sparse hull. Fractional coordinate sections of a 0/1 polytope need not themselves be integral. Large coefficients for general 0/1 polytopes and ill-conditioned 0/1 matrices are well-established phenomena; see, for example, Alon and Vu [2]. The restriction here is to selected flow–simplex products with only two explicit states. The lower bound concerns coefficient *magnitudes*; their bit lengths can remain polynomial. It does not imply separation hardness, large extension complexity, or a corresponding statement on planar networks.

8 Sparse coefficient growth on series–parallel networks

The preceding construction uses unrestricted bipartite layers. A more explicit obstruction occurs on a much narrower graph family, provided the number of simplex states is allowed to grow. For $L \geq 1$, let G_L have vertices $v_0, \dots, v_L$, two arcs $a_i, b_i : v_{i-1} \rightarrow v_i$ for each $i \in [L]$, and a bypass $h : v_0 \rightarrow v_L$. All capacities and the source–sink flow value are one. This is a directed two-terminal series–parallel graph: first put each pair in parallel, compose the pairs in series, and put their chain in parallel with h . In a disaggregation, conservation forces

$$f_{a_i}^j + f_{b_i}^j = w_j \quad (i \in [L]), \quad f_h^j = \lambda_j - w_j, \quad 0 \leq w_j \leq \lambda_j. \quad (91)$$

The vector w is the *shared branch profile*. Its total is $\sum_j w_j = 1 - x_h$. Keeping only this total can lose essential compatibility between the serial gadgets.

8.1 A balanced-incidence construction

Lemma 8.1 (Balanced incidence gives a coordinate-section edge). *Let $K \in \{0, 1\}^{N \times N}$ be invertible, with every row nonempty and proper. Suppose $\alpha \in \mathbb{Q}_{>0}^N$ and $\beta > 0$ satisfy $K^\top \alpha = \beta \mathbf{1}$. On G_N there is a sparse observation pattern with $m = N$ such that, for any two distinct rows r, s , a coordinate section has locally the edge*

$$\alpha_r(U - c) + \alpha_s(V - c) \geq 0, \quad c = 1/(8N). \quad (92)$$

Here U, V are individual observed products, one from each row; every other original coordinate is fixed. Consequently the hull has a relative facet with product-coefficient ratio α_r/α_s , invariant under affine-hull equations.

Proof. Put $a = 1/(2N)$ and let $S_i = \{j : K_{ij} = 1\}$. Fix $y_j = 2a = 1/N$ for $j \in [N]$, so the residual state is zero. Observe (a_i, j) precisely when $j \notin S_i$. Since each S_i is proper, choose one such cell in row r and one in row s as U, V . Fix all other observations to $c = a/4$, and fix

$$x_h = 1/2, \quad x_{a_i} = |S_i|a + (N - |S_i|)c, \quad x_{b_i} = 1/2 - x_{a_i}. \quad (93)$$

All these arc flows are strictly between zero and $1/2$, except for the stated bypass value. Write $\delta_r = U - c$, $\delta_s = V - c$, and $\delta_i = 0$ in other rows. Since each unobserved a_i entry is at most its branch profile, every feasible state decomposition obeys

$$K(w - a\mathbf{1}) \geq -\delta, \quad \mathbf{1}^\top w = 1/2.$$

Multiplication by $\alpha^\top$ proves $\alpha^\top \delta \geq 0$, which is (92).

We prove the converse in an open neighborhood of $(U, V) = (c, c)$. Put $R = \alpha_r/\alpha_s$, let τ be zero except for $\tau_s = R\delta_r + \delta_s$, and define

$$d' = K^{-1}(-\delta + \tau), \quad w = a\mathbf{1} + d'. \quad (94)$$

On the proposed feasible side, $\tau_s \geq 0$. Also $\beta \mathbf{1}^\top d' = \alpha^\top (-\delta + \tau) = 0$, so the profile sums to $1/2$. For example, with $K_* := \|K^{-1}\|_\infty$, the neighborhood

$$|\delta_r|, |\delta_s| < \varepsilon := \frac{a}{16(1+R)(1+K_*)}$$

ensures $\|d'\|_\infty < a/16$ and $0 \leq \tau_s < a/16$. Indeed $-\delta + \tau$ has only entries $-\delta_r$ and $R\delta_r$. Every observed value then lies strictly between zero and $a/2$, whereas each w_j lies between $15a/16$ and $17a/16$.

Initially set $f_{a_i}^j = w_j$ for $j \in S_i$ and set the other entries to their prescribed observations. The aggregate of row i is $x_{a_i} + \tau_i$, by (94). In row s reduce one entry with $j \in S_s$ by τ_s ; such a state exists and its entry remains nonnegative. All a_i entries are now in $[0, w_j]$ and have the required aggregates. Set $f_{b_i}^j = w_j - f_{a_i}^j$ and $f_h^j = 2a - w_j$. These are nonnegative, at most $2a$, and satisfy every state balance, aggregate, and observation. The residual flow is zero. Disaggregation proves sufficiency, including points on the boundary line. The local section has two-dimensional interior, so lemma 6.7 gives the final claim. $\square$

Example 8.2 (Five vertices and nine arcs). Take $N = 4$ and

$$S_1 = \{1, 2, 3\}, \quad S_2 = \{1, 4\}, \quad S_3 = \{2, 4\}, \quad S_4 = \{3, 4\}.$$

The incidence matrix is invertible: its last three homogeneous row equations give $d'_1 = d'_2 = d'_3 = -d'_4$, and its first gives $3d'_4 = 0$. The balancing weights are $(2, 1, 1, 1)$ with $\beta = 3$. There are seven observations on G_4 , which has five vertices, nine arcs, and cycle rank five. With $U = z_{a_1, 4}$ and $V = z_{a_2, 2}$, the section edge is $2U + V \geq 3/32$. Here $x_{a_1} = 13/32$, $x_{a_i} = 5/16$ for $i = 2, 3, 4$, $x_h = 1/2$, and each explicit weight is $1/4$. More generally, use $N = k + 1$, $S_1 = [k]$, and $S_{i+1} = \{i, k + 1\}$ for $i \in [k]$, with $k \geq 3$. The same kernel argument proves invertibility and the weights $(k - 1, 1, \dots, 1)$ balance every column to k . This gives ratio $M = k - 1$, with $M + 3$ vertices, $2M + 5$ arcs, $M + 2$ explicit states, and $1 + M(M + 1)$ observations. The next construction attains exponential ratios with only linearly many observations.

Remark 8.3 (Why separate scalar composition fails). In lemma 8.1, put $U = c - \eta$, $V = c$ for sufficiently small $\eta > 0$. The joint section is infeasible because $\alpha^\top \delta = -\alpha_r \eta < 0$. Nevertheless, each serial gadget is individually feasible with total branch flow $1/2$ and the same prescribed state weights. In row r , start from the nominal profile $a\mathbf{1}$ and move η of profile mass from a state outside S_r to a state in S_r . Keep the prescribed observed entries, and fill the entries in S_r to their new profiles. The observed row total decreases by η and the unobserved total increases by η , so the aggregate is unchanged. All bounds remain valid for small η , including if the donor is the free observed state. Every other gadget uses the nominal profile. Thus independently eliminating profiles and passing only their total between gadgets accepts this infeasible point. The missing information is a single *common* profile, not a scalar flow balance.

8.2 A sparse Fibonacci family

Theorem 8.4 (Exponential ratios on a flat series-parallel chain). *For every $q \geq 3$ there is an instance on G_{2q-1} with $m = 2q - 1$ and $5q - 4$ observed products such that every finite original-coordinate inequality description contains product coefficients in ratio F_q , where $F_1 = F_2 = 1$ and $F_i = F_{i-1} + F_{i-2}$. The graph has $2q$ vertices, $4q - 1$ arcs, and cycle rank $2q$. It is planar and has treewidth two. The original sparse input has encoding length $O(q \log q)$ and only numerical data in $\{-1, 0, 1\}$.*

Proof. Put $N = 2q - 1$ and form a square 0/1 incidence matrix D . Its primary rows are $P_1, \dots, P_q$ and its auxiliary rows are $H_0, H_3, \dots, H_q$. Its columns are the following subsets of the row set, in the displayed order:

$$\{H_0, P_1\}, \{H_0, P_2\}; \quad \{H_i, P_i\}, \{H_i, P_{i-1}, P_{i-2}\} (3 \leq i \leq q); \quad \{P_q, P_{q-1}\}. \quad (95)$$

There are $2 + 2(q - 2) + 1 = N$ columns and $4 + 5(q - 2) + 2 = 5q - 4$ ones. Each row has two or three ones and each column has two or three ones, so every row and column is nonempty and proper. Set $\gamma = F_{q+1}$ and

$$\alpha_{P_i} = F_i, \quad \alpha_{H_0} = \gamma - 1, \quad \alpha_{H_i} = \gamma - F_i \quad (3 \leq i \leq q).$$

All weights are positive. The recurrence and the last column show $D^\top \alpha = \gamma \mathbf{1}$.

To prove invertibility, suppose $D^\top \eta = 0$. Subtracting the first pair of column equations gives $\eta_{P_1} = \eta_{P_2}$; subtracting each subsequent pair gives $\eta_{P_i} = \eta_{P_{i-1}} + \eta_{P_{i-2}}$. Thus $\eta_{P_i} = F_i \eta_{P_1}$. The last column then gives $F_{q+1} \eta_{P_1} = 0$. Every primary and auxiliary entry is zero.

We use the complement $K = \mathbf{1}\mathbf{1}^\top - D$ as the allowed-set matrix in [lemma 8.1](#); consequently its *observed* cells are precisely the sparse ones of D . Put $\sigma = \mathbf{1}^\top \alpha$. Summing the weights gives $\sigma = (q - 1)\gamma + 1$, whence

$$K^\top \alpha = (\sigma - \gamma) \mathbf{1}, \quad \sigma - \gamma = (q - 2)\gamma + 1 > 0.$$

The complement is invertible. Indeed, if $K^\top \eta = 0$, then $D^\top \eta = (\mathbf{1}^\top \eta) \mathbf{1}$, so $\eta = (\mathbf{1}^\top \eta) \alpha / \gamma$. Summing this identity and using $\sigma > \gamma$ forces $\mathbf{1}^\top \eta = 0$ and then $\eta = 0$. Its rows are nonempty and proper as required.

Choose U in row P_q and the last column of (95), and V in row P_1 and the first column. These cells are observed. The exact local section is

$$F_q U + V \geq (F_q + 1)c, \quad c = 1/(8N). \quad (96)$$

The facet conclusion follows from [lemma 8.1](#). The graph and observation counts follow from the definition of G_N . Its underlying simple graph is a cycle. Explicit bags $\{v_0, v_i, v_{i+1}\}$ for $0 \leq i < N$, in their natural path order, give a tree decomposition of width two; for $N \geq 3$ the cycle requires that width. Parallel arcs do not affect treewidth or planarity. All graph, state, and observation lists have $O(q)$ entries, with indices of $O(\log q)$ bits. The model itself uses only unit data; even the fixed section coordinates in (93) have $O(\log q)$ bits. Since F_q grows exponentially, the required integer coefficient magnitudes are superpolynomial in the input length, while their bit lengths need only grow linearly in q . $\square$

The matrix D itself also satisfies [lemma 8.1](#) and gives the ratio F_q , by choosing any observed cell in each of rows P_q, P_1 . That version uses $N^2 - (5q - 4)$ observations. Passing to its invertible complement preserves the balancing weights and makes the observation set sparse.

Corollary 8.5 (Simple graphs of maximum degree three). *The same ratio F_q can be realized on a simple acyclic directed two-terminal series-parallel unit-flow network of maximum undirected degree three, with $6q - 3$ vertices, $8q - 4$ arcs, unchanged cycle rank $2q$, and the same states and observed products.*

Proof. In G_N , split each of the $N - 1$ internal joins into an incoming vertex and an outgoing vertex connected by a new forward arc. Subdivide each b_i once. The resulting graph has $3N$ vertices and $4N$ arcs. The subdivisions remove the parallel pairs; terminals and split joins have degree three, and subdivision vertices have degree two. All arcs point forward. The construction remains two-terminal series-parallel and planar. Such graphs have treewidth at most two: inductively keep a bag containing the two terminals, join two such bags for parallel composition, and add the bag $\{s, v, t\}$ for a series composition through v .

Every old flow has a unique extension: each connector carries the chain flow and each subdivided arc carries the old b_i flow. Conversely contraction recovers the old flow. The new unit capacities are

redundant because these are nonnegative unit flows on an acyclic network. The same statements hold separately for every state flow. All observed a_i arcs remain unchanged. In the coordinate section, fix each connector to $1/2$ and each new subdivision coordinate to the corresponding old aggregate. Thus precisely the same free (U, V) section is obtained, and [lemma 6.7](#) preserves its facet-ratio conclusion. Substituting $N = 2q - 1$ gives the counts. $\square$

The hulls in this section are again 0/1 polytopes, by the path argument following [corollary 7.2](#). The obstruction does not conflict with integrality results for aggregate multiflows on series-parallel digraphs. Almoghrabi et al. [[1](#), Remark 1] explicitly distinguish aggregate arc flows from the full commodity-flow vector. Here prescribed entries of that state vector carry the obstruction. The exponential construction lets both block cycle rank and the number of states grow; it does not disprove a bound at fixed state count. We next show that a fixed state count does control this flat topology.

9 Exact hulls for chains with few states

For the same graph G_L , arbitrary observations on either gadget arc and on the bypass admit a small common-profile description. Eliminating the residual profile coordinate sharpens the general determinant bound and gives a unit-coefficient original hull when $m \leq 3$.

9.1 The complete shared-profile system

First check $x \in P$, $y \in \Delta_m$, and $z \geq 0$ on the observed coordinates. These conditions are part of every description below. Put $t = 1 - x_h$, and index all $d = m + 1$ states by $\{0, \dots, m\}$. For each gadget i , partition the states into A_i, B_i, T_i, U_i : only a_i is observed, only b_i is observed, both are observed, or neither is observed, respectively. In particular $0 \in U_i$. Write $u_{ij} = z_{a_i,j}$ and $v_{ij} = z_{b_i,j}$ wherever present, and define

$$R_i = x_{a_i} - \sum_{j \in A_i \cup T_i} u_{ij} + \sum_{j \in B_i} v_{ij}. \quad (97)$$

The use of T_i for a state class is unrelated to the table polytope $\mathcal{T}$ in [section 7](#).

Proposition 9.1 (Exact profile formulation). *Subject to the preceding original-domain checks, $(x, y, z) \in H$ if and only if there is $w \in \mathbb{R}^{m+1}$ satisfying*

$$0 \leq w_j \leq \lambda_j \quad (j \in \{0, \dots, m\}), \quad \sum_{j \in \{0, \dots, m\}} w_j = t, \quad (98)$$

$$w_j = \lambda_j - z_{h,j} \quad ((h, j) \in \mathcal{O}), \quad (99)$$

$$w_j \geq u_{ij} \quad (j \in A_i),$$

$$w_j \geq v_{ij} \quad (j \in B_i),$$

$$w_j = u_{ij} + v_{ij} \quad (j \in T_i), \quad (100)$$

$$\sum_{j \in B_i} w_j \leq R_i \leq \sum_{j \in B_i \cup U_i} w_j \quad (i \in [L]). \quad (101)$$

Proof. Every state disaggregation supplies its profile through (91). Its observed entries give (99)–(100). In gadget i , the a_i entries are u_{ij} in $A_i \cup T_i$, are $w_j - v_{ij}$ in B_i , and are free in $[0, w_j]$ in U_i . Thus their possible aggregate is exactly the interval

$$\left[\sum_{A_i \cup T_i} u_{ij} + \sum_{B_i} (w_j - v_{ij}), \quad \sum_{A_i \cup T_i} u_{ij} + \sum_{B_i} (w_j - v_{ij}) + \sum_{U_i} w_j \right].$$

Membership of x_{a_i} in this interval is (101). Conversely these inequalities allow the free U_i entries to be chosen with the required sum, independently for each gadget. All entries lie in $[0, w_j]$: for a doubly observed state this follows from $w_j = u_{ij} + v_{ij}$ and the original nonnegativity check. Set every b_i entry to $w_j - f_{a_i}^j$ and the bypass entry to $\lambda_j - w_j$. State capacities and balances hold, and $x_{a_i} + x_{b_i} = t$ supplies each b_i aggregate. The bypass aggregate is $\sum_j \lambda_j - t = x_h$. This is a complete disaggregation. If a weight is zero, its profile and all its entries are zero, so no positive-weight assumption has been used. $\square$

Eliminate the unobserved residual coordinate by $w_0 = t - \sum_{j=1}^m w_j$, and write $\hat{w} = (w_1, \dots, w_m)$. The residual bounds become

$$\mathbf{1}^\top \hat{w} \leq t, \quad -\mathbf{1}^\top \hat{w} \leq \lambda_0 - t. \quad (102)$$

Since $0 \in U_i$, the endpoint rows become

$$\mathbf{1}_{B_i}^\top \hat{w} \leq R_i, \quad \mathbf{1}_{A_i \cup T_i}^\top \hat{w} \leq t - R_i. \quad (103)$$

All remaining rows have singleton normals $\pm e_j$. Thus each nonzero normal belongs to the fixed universe

$$\mathcal{N}_m = \{\mathbf{1}_S : \emptyset \neq S \subseteq [m]\} \cup \{-e_j : j \in [m]\} \cup \{-\mathbf{1}\}. \quad (104)$$

Repeated normals represent different affine right-hand sides; keep the minimum of these right-hand sides at a query point. A zero normal gives a scalar inequality and must be checked separately. Every original flow or product has coefficient in $\{-1, 0, 1\}$ in every individual right-hand side, including $t - R_i$.

9.2 Finite circuits and constructive recovery

Let Δ_m^{01} be the maximum absolute determinant of a square 0/1 matrix of order at most m , including the empty determinant one. In particular $\Delta_1^{01} = \Delta_2^{01} = 1$, $\Delta_3^{01} = 2$, and $\Delta_m^{01} \leq \max(1, m^{m/2})$ by Hadamard's inequality.

Theorem 9.2 (Fixed-state chain hull and recovery). *For $m \geq 1$, the sparse hull on G_L has a finite original-coordinate description with integer flow and product coefficients of magnitude at most $(m+1)\Delta_m^{01}$. After $2^{O(m^2)}$ preprocessing depending only on m , exact separation and state-flow recovery each require*

$$O(mL + |\mathcal{O}| + m) + 2^{O(m^2)}$$

rational arithmetic operations. Recovery calls no online linear program. Rational encoding lengths remain polynomial in the input length. For $m = 0$, the hull is simply the original flow polytope.

Proof. Form the reduced profile inequalities $a^\top \hat{w} \leq \gamma_a$ for the distinct present nonzero normals, where γ_a is the minimum of all right-hand sides with normal a . There are at most $2^m + m$ such normals. The system is feasible exactly when all its nonnegative cancellation certificates have nonnegative right-hand side, by Farkas' lemma. The cone of such certificates is

$$\{\mu \geq 0 : \sum_a \mu_a a = 0\}.$$

It is pointed. Each extreme ray has minimally dependent support, of size at most $m+1$; otherwise a linear dependence within its support permits a small perturbation in both directions and contradicts extremality. Conversely every minimal positive dependence gives such an extreme ray. We call these supports, with their primitive positive integer weights, *positive circuits*.

For a support of size s , select $s - 1$ independent columns in its row matrix. The signed $(s - 1)$ -row minors give a nonzero kernel vector and hence its circuit weights after a common sign change and division by the greatest common divisor. Each row is a 0/1 vector or its negative. Changing whole-row signs bounds every such minor by Δ_m^{01} . Therefore every primitive weight is at most Δ_m^{01} . The extreme rays generate the certificate cone, so the finitely many tests

$$\sum_{a \in C} \mu_a \gamma_a \geq 0 \quad \text{for every positive circuit } C \text{ of present normals} \quad (105)$$

are necessary and sufficient, together with the scalar zero-row checks. Each circuit is one in the full fixed universe (104).

Expanding (105) over all choices of an original row for each participating normal gives a finite affine description. Positivity of the weights makes this expansion exact: the weighted sum of minima is the minimum over independent choices. Each affine inequality sums at most $m + 1$ rows, with weights at most Δ_m^{01} , and hence has the claimed original-coordinate coefficient bound. At an infeasible query it suffices to output the rows attaining the minima in one violated circuit. This produces a globally valid affine inequality, even though other rows may attain the minima at other points. The original-domain prechecks have unit coefficients as well.

Enumerate all supports of size at most $m + 1$ in the fixed universe, and use exact elimination to keep their positive circuits. There are $2^{O(m^2)}$ candidates. Constructing and grouping the profile rows takes $O(mL + |\mathcal{O}| + m)$ arithmetic operations; the circuit tests then have parameter-only cost $2^{O(m^2)}$. For this count, preindex subset normals by bit masks and cache the singleton indices. A scan of each gadget builds its two subset masks and right-hand sides, while each local cell row uses a cached index; one need not copy an m -entry normal for every local row. The masks have $O(m)$ bits, which is included in the polynomial bit-length qualification below.

For recovery, also precompute inverses of all nonsingular matrices formed by m distinct normals of $\mathcal{N}_m$. At a feasible query enumerate those whose rows are present and solve their equalities against the corresponding grouped right-hand sides. Test each candidate against all the grouped inequalities, and keep the first feasible candidate. This succeeds: the profile polyhedron is nonempty and bounded by $0 \leq w_j \leq \lambda_j$, so has a vertex; at a vertex the active normals span $\mathbb{R}^m$, even if the polyhedron is lower dimensional. If they failed to span, a sufficiently small displacement in a nonzero orthogonal direction and its negative would remain feasible, contradicting that it is a vertex. Thus the enumeration includes that vertex. Its number of candidates and all candidate tests are bounded by $2^{O(m^2)}$.

Set $w_0 = t - \sum_{j=1}^m w_j$. In each gadget, the remaining U_i entries must sum to $R_i - \sum_{B_i} w_j \in [0, \sum_{U_i} w_j]$. Fill these entries in any fixed order, repeatedly taking the smaller of the remaining sum and w_j . This takes $O((m + 1)L)$ arithmetic operations and gives the disaggregation in [proposition 9.1](#). Divide positive-weight states by their weights to obtain at most $m + 1$ graph points, omitting zero states. Writing all state-arc entries has this same $O((m + 1)L)$ output scale.

All bases have integer entries of magnitude at most one. Their inverses have cofactors bounded by Δ_m^{01} and nonzero integer determinants bounded by Δ_m^{01} . Grouping selects input-derived affine right-hand sides; circuit tests and inverse-basis recovery therefore use numbers with polynomial encoding length. Greedy filling uses only additions, comparisons, and minima of these rational values. Normalization by positive rational weights preserves polynomial encoding length. The arithmetic parameter dependence is exponential, but no unbounded-precision unit-cost assumption is needed for the bit-length assertion. $\square$

Remark 9.3 (The unreduced signed-subset library). Keeping all $d = m + 1$ profile coordinates yields normals $\{\pm \mathbf{1}_S : \emptyset \neq S \subseteq \{0, \dots, m\}\}$. The same proof gives the bound $(d + 1)\Delta_d^{01}$ and $2^{O(d^2)}$

parameter work. Its positive-circuit counts for $d = 1, 2, 3$ are respectively 1, 5, 41, with largest primitive weights 1, 1, 2. These are counts of the full signed-subset universe, not of the rows that necessarily occur in a particular instance. Eliminating the residual coordinate gives the smaller universe (104): for $m = 1, 2, 3$ it has 2, 6, 11 normals and 1, 5, 16 positive circuits, respectively, again with largest weights 1, 1, 2. The stated finite enumeration reproduces these counts exactly; they are not assumptions in the proof. In particular the 41-circuit library is unnecessary for a chain with two explicit states.

9.3 Two explicit states: five tests and unit coefficients

For $m = 2$, the reduced normal universe is just $\{\pm e_1, \pm e_2, \pm(e_1 + e_2)\}$. After grouping, the profile region takes the form

$$\ell_1 \leq w_1 \leq u_1, \quad \ell_2 \leq w_2 \leq u_2, \quad \ell_s \leq w_1 + w_2 \leq u_s. \quad (106)$$

All six endpoint groups are present because of the explicit and residual weight bounds. This is the same three-direction geometry used in the theta oracle, but here the individual profile region itself describes all serial gadgets together.

Theorem 9.4 (Unit coefficients and a five-test chain oracle). *On G_L with at most two explicit simplex variables and arbitrary observations, the sparse hull has a finite exact description with all flow and product coefficients in $\{-1, 0, 1\}$. With two explicit variables, after the original-domain and zero-row checks, membership is equivalent to the five conditions*

$$\ell_1 \leq u_1, \quad \ell_2 \leq u_2, \quad \ell_s \leq u_s, \quad \ell_1 + \ell_2 \leq u_s, \quad \ell_s \leq u_1 + u_2. \quad (107)$$

Exact separation and constructive decomposition take $O(L + |\mathcal{O}| + 1)$ rational arithmetic operations, with no circuit enumeration and no online linear program.

Proof. The possible sum of two interval variables is $[\ell_1 + \ell_2, u_1 + u_2]$. Thus (107) are exactly the conditions that both intervals are nonempty and this sum interval meets $[\ell_s, u_s]$. On a feasible query set

$$s_* = \max(\ell_s, \ell_1 + \ell_2), \quad w_1 = \max(\ell_1, s_* - u_2), \quad w_2 = s_* - w_1.$$

The tests ensure $s_* \leq \min(u_s, u_1 + u_2)$, and these choices satisfy every bound in (106). Complete the flows by the greedy construction in theorem 9.2.

We must check original coefficients, since unit circuit weights alone would not exclude repeated occurrences of the same product. The five positive circuits are the three opposite pairs and

$$\{e_1, e_2, -\mathbf{1}\}, \quad \{-e_1, -e_2, \mathbf{1}\}, \quad (108)$$

all with weights one. The following occurrence analysis applies to any affine row selected in each normal group. No circuit contains both $-e_j$ and $+e_{3-j}$, or both $+e_j$ and $+\mathbf{1}$.

For an a_i product in an only- a_i state j , its negative occurrences are in the local row $-e_j$ and the endpoint row $+\mathbf{1}_{B_i}$; since $j \notin B_i$, the latter is either zero or $+e_{3-j}$. Its only positive occurrence is in the other endpoint row $+\mathbf{1}_{A_i \cup T_i}$. Thus a circuit cannot select two negative occurrences. If state j is doubly observed, the positive local row $+e_j$ is also available. The other positive normal contains j , so is either the same normal $+e_j$ or $+\mathbf{1}$; one row is selected per normal, and these two distinct normals do not occur together in a circuit. The negative occurrences remain as above. Hence every a_i product has coefficient of magnitude at most one.

For a b_i product in an only- b_i state j , the negative occurrences are $-e_j$ and $+1_{A_i \cup T_i}$, whose latter normal excludes j ; the positive occurrence is the B_i endpoint. The same argument applies. A doubly observed b_i product occurs only in the two opposite local singleton rows, with opposite signs, because R_i uses the observed a_i entry in that state. A bypass product likewise occurs only in two opposite singleton rows. Each zero-row inequality separately has unit coefficients.

The coordinate x_{a_i} occurs only in its two endpoint rows, once with each sign, and x_{b_i} occurs only in the original flow precheck. The shared coordinate x_h needs an additional check. In a positive singleton row its coefficient is zero or -1 , in a negative singleton row it is zero, in the positive full row it is zero or -1 , and in the negative full row it is always $+1$. These statements follow by inspecting (102) and (103); for example a full B_i endpoint has zero x_h coefficient, whereas a full $A_i \cup T_i$ endpoint has coefficient -1 . For the first circuit in (108), two negative contributions from its positive singleton rows are offset by the compulsory $+1$ from the negative full row. All opposite pairs and the second circuit also have coefficient magnitude at most one. Consequently every affine branch of the five tests is a unit flow/product inequality, proving the claimed finite description.

For $m = 1$ the reduced system is an interval. Alternatively, embed it as the coordinate section $y_2 = 0$ of the two-state instance with no observations in label 2. Disaggregation forces state 2 to zero, so restriction of the established unit description is exact and preserves its flow/product coefficients. The case $m = 0$ is the original flow description. Grouping rows, checking the five tests, and filling the state flows have the asserted linear cost. $\square$

9.4 Three explicit states and a sharp state-count threshold

The coefficient conclusion extends one state further. In this case some direct circuit sums have bypass-flow coefficient two, but a single original flow equation removes it. This is a valid change of an ambient inequality representative, in contrast to changing either free coefficient in the coordinate sections of lemma 6.7.

Theorem 9.5 (Sharp unit-coefficient threshold for flat chains). *For arbitrary observations on G_L , three explicit simplex variables still admit a finite exact original-coordinate description with unit flow and product coefficients. At most sixteen fixed positive-circuit tests, with the original-domain and zero-row checks, give exact separation and constructive recovery in $O(L + |\mathcal{O}| + 1)$ rational arithmetic operations. Four explicit variables already allow a hull with no such unit-coefficient description.*

Proof. For $m = 3$, the positive circuits of (104) have the following four forms. Each listed normal has weight one unless specified otherwise:

$$\{1_S\} \cup \{-e_j : j \in S\} \quad (\emptyset \neq S \subseteq [3]); \quad (109)$$

$$\{-1\} \cup \{1_S : S \in \pi\} \quad (\pi \text{ a partition of } [3]); \quad (110)$$

$$\{-1, -e_i, e_i + e_j, e_i + e_k\} \quad (\{i, j, k\} = [3]); \quad (111)$$

$$\{-1, e_1 + e_2, e_1 + e_3, e_2 + e_3\}, \quad \text{weight two on } -1. \quad (112)$$

Their numbers are 7, 5, 3, 1. Here is a direct completeness check. Without -1 , a positive dependence decomposes into the dependencies of each positive subset on its negative singleton coordinates, so minimality leaves exactly (109). With -1 , presence of $+1$ forces the opposite pair. Otherwise there are at most three other normals, by the circuit support bound. Two negative singleton normals would leave at most one positive proper subset, which cannot cover all three coordinates and also supply an excess in either negative singleton coordinate. With one negative singleton $-e_i$, two positive subsets are necessary. The opposite singleton $+e_i$ is excluded by minimality. The coordinate equations then force the positive subsets to be $\{i, j\}$ and $\{i, k\}$, giving (111). With no negative

singleton, the positive subsets form a minimal uniformly weighted cover. If a singleton is present, overlapping pairs would force one of their other coordinates to exceed the common cover value, so the sets form a partition. With only pairs, all three pairs are needed, and their equal weights give (112). This also verifies positivity and minimality of the displayed list.

All positive-normal weights are one. No circuit contains both $-e_j$ and a positive subset normal excluding j , or both $+e_j$ and a different positive subset normal containing j . These facts follow immediately from the four forms. The occurrence argument in [theorem 9.4](#) therefore proves that every product coefficient has magnitude at most one for every choice of affine branches. The only doubled circuit weight belongs to $-\mathbf{1}$, whose right-hand side $\lambda_0 - t$ has no product coordinate. Each x_{a_i} has at most its two opposite-signed endpoint occurrences, each with weight one, and no x_{b_i} occurs in a profile right-hand side.

It remains to handle x_h . Negative singleton rows have coefficient zero; the negative full row has coefficient $+1$. A positive subset row has coefficient zero or -1 . Thus circuits (109) and (111) have unit x_h coefficients. In (110), the only exception is coefficient -2 : the partition must consist of three singletons and all three chosen right-hand sides must be upper endpoints $t - R_i$. They belong to distinct gadgets, since each gadget supplies only one upper endpoint. Choose one of them. Its x_{a_i} coefficient is -1 , with no other contribution from that gadget. In the circuit inequality written as an affine expression at least zero, add the identically zero flow expression

$$x_{a_i} + x_{b_i} + x_h - 1.$$

The coefficients of x_h, x_{a_i}, x_{b_i} become $-1, 0, 1$.

In (112), the only exception is coefficient $+2$, occurring when all three positive pair rows have zero x_h coefficient. A positive pair row can only come from a gadget endpoint, so these are three lower endpoints R_i from distinct gadgets. Choose one and subtract its flow expression $x_{a_i} + x_{b_i} + x_h - 1$. The three affected coefficients become $1, 0, -1$. No other coefficient changes. These replacements preserve validity on the original flow domain and leave the violation at any query that passes the flow precheck unchanged. Hence every branch of the exact circuit description has a unit-coefficient representative.

The circuit and inverse-basis libraries now have fixed size. [Theorem 9.2](#) supplies linear-cost separation and recovery, including zero weights and lower-dimensional profile regions. Finally [example 8.2](#) has four explicit states and a local section normal $(2, 1)$. By [lemma 6.7](#), this ratio cannot be removed by any affine-hull equation, so no unit-coefficient description exists. Additional unused explicit variables can be fixed to zero if a counterexample with more than four is desired. $\square$

Example 9.6 (Both bypass-coefficient repairs are needed). Take $L = 3$, $x_h = 1/2$, and first let $y_1 = y_2 = y_3 = 1/3$, $x_{a_i} = 2/5$, $x_{b_i} = 1/10$. Observe only (a_i, i) , with all three products zero. The singleton-partition certificate is

$$\sum_{i=1}^3 (t - R_i) + (\lambda_0 - t) = 2t - \sum_{i=1}^3 x_{a_i} + \sum_{i=1}^3 z_{a_i, i} + \lambda_0 = -1/5.$$

It has x_h coefficient -2 ; adding any selected gadget balance gives a unit-coefficient violated inequality. For the other repair, take $y_1 = y_2 = y_3 = 1/4$, $x_{a_i} = 1/20$, $x_{b_i} = 9/20$. Observe only b_i products, on state pairs $\{1, 2\}, \{1, 3\}, \{2, 3\}$ in gadgets 1, 2, 3, respectively, each with value $1/20$. Now $R_i = 3/20$ and the pair-triangle certificate is

$$R_1 + R_2 + R_3 + 2(\lambda_0 - t) = -1/20.$$

Its x_h coefficient is $+2$; subtracting a selected gadget balance gives the unit form. Both query points satisfy all original flow, simplex, and separate McCormick inequalities. The exact joint cuts detect the missing common profile.

Corollary 9.7 (Only observed labels count). *Let $J = \{j : (e, j) \in \mathcal{O} \text{ for some } e\}$ and $a = |J|$. All fixed-state chain conclusions apply with a in place of m . In particular, at most three observed labels guarantee unit flow/product coefficients, regardless of the number of unobserved simplex labels; at most two observed labels permit the five-test oracle. General separation and compact recovery take $O((a+1)L + |\mathcal{O}| + m) + 2^{O(a^2)}$ rational arithmetic operations.*

Proof. Apply the exact state merger in [theorem 3.2](#), retaining the labels in J and merging all others with the residual state. The merged weight is $1 - \sum_{j \in J} y_j$. Keep the original simplex constraints on all m coordinates and apply the preceding theorems to this reduced state list. Substituting the merged weight changes no flow or product coefficient. The proportional refinement [\(19\)](#) recovers each unused label from the same normalized merged flow. It can be stored as one flow and its constituent weights, so the stated compact-recovery cost includes only $O(m)$ additional bookkeeping. Writing every entry of all global state flows instead can require $\Theta((m+1)|E|)$ output operations. If $a = 0$, the hull is simply $P \times \Delta_m$ and no profile elimination is needed. $\square$

These theorems are specific to a chain of parallel pairs with one bypass. A general nested series-parallel construction can require several interacting profiles. Neither the five-test result nor the fixed-state determinant bound above asserts a description for that larger class. Conversely, [theorem 8.4](#) shows why a coefficient bound independent of the state count is impossible even on this flat family.

10 Implementation, verification, and computational comparisons

The implementations serve two distinct purposes: constructing small numerical LPs and producing exact rational hull certificates. The experiments compare both with strengthened versions of classical disaggregation. They show large model-size savings, but also cases where a simpler LP is faster and cases where global state merging explains most of an apparent compression advantage.

10.1 Implemented scope and certificate semantics

[Table 1](#) distinguishes the three usable interfaces from the bounded-rank verification code. The specialized separators use Python’s standard library and rational arithmetic throughout. They check original flow and simplex constraints, retain the original affine expressions of active bounds, and return a cut with strictly positive exact violation or a decomposition whose positive-weight normalized flows belong to P . The graph separator rejects unsupported blocks explicitly. The flat oracle merges unobserved labels, eliminates the residual profile, and uses the five-test formula when at most two labels are observed. At three labels it uses the 16 circuits and both flow-balance repairs of [theorem 9.5](#) for separation. Recovery uses finite inverse bases at three or more observed labels. Its stored default is shared across unused global labels. Compatibility accessors can materialize the former dense state arrays, with the corresponding output cost. Tuple normal construction and hashing retain a dimension factor relative to the preindexed-mask operation count in [theorem 9.2](#); apart from the constructor’s $O(|\mathcal{O}| \log |\mathcal{O}|)$ observation sorting, the implemented cost is linear in L at fixed observed-label count.

The general formulation stores exact rational coefficients, including the observed-coordinate substitutions, but calls a numerical LP solver. Its phase-I certificate routine rationalizes dual

Table 1: Implemented scope. Exact assembly of an LP does not make its floating-point optimizer an exact certificate.

Component	Supported input	Returned evidence
Parallel-path separator	Arbitrarily oriented multigraphs whose cyclic blocks are parallel paths; loops and bridges included	Exact original-coordinate cut or compact rational decomposition
Flat-chain separator	Canonical unit-data G_L , arbitrary observations; parameter is the number of observed labels	Exact cut or compact decomposition; unit flow/product coefficients through three observed labels
General compressed EF	Arbitrary bounded equality-flow graphs; initial or observed-coordinate-eliminated formulation	Exact rational model; numerical LP solution; optionally an exactly checked Farkas cut
Bounded-rank libraries	Supplied small signed-normal systems and structural verification instances	Verification prototype for support and recovery; not a complete production general-graph oracle

weights and, if necessary, repairs their supported kernel equations by exact elimination. A returned cut is accepted only after checking nonnegative weights, exact cancellation of every auxiliary coordinate, and strictly positive rational violation. The API distinguishes `certified_outside`, `numerically_feasible`, `uncertified_outside`, and `solver_failure`. The second status is not exact membership, and failure of exact multiplier recovery is not an infeasibility certificate. This is an implementation of classical Farkas projection, not a new generic separation theorem.

Code indices are zero based, with explicit states $0, \dots, m-1$ and residual state m ; this is only an indexing change from the manuscript. The LP baseline generator uses outgoing-minus-incoming balances, and its adapter negates them before calling either production model. Float inputs to exact interfaces mean their decimal text. The optimization experiment uses a small-denominator reconstruction adapter, requires error below 10^{-7} and exact aggregate balances, and then checks the returned decomposition exactly. This adapter is suitable for the stated synthetic instances; it is not a general certified rational LP optimizer.

The verification suite compares against independently assembled full-state LPs and, for small graphs, directly enumerated path-simplex vertices. The updated flat tests include arbitrary observations and zero weights, both three-label repairs with exact global cut checks, the four-label ratio-two obstruction, and sparse observed labels among many original simplex coordinates. They also test an interior-residual point with zero explicit profile and positive residual branch flow: this rules out incorrectly reusing a basis construction that forces the explicit profile sum to the full branch flow. Separate tests check every new rank-three inverse basis. The unchanged independent flat audit verifies 192 exact decompositions and 708 exact violated cuts, with 233 numerical path-hull comparisons. The general compressed audit passes 600 objective comparisons, 758 membership comparisons, and 476 exactly checked Farkas cuts, each also tested by numerical full-hull support optimization. Its structural integration checks recover the coefficient ratios two, five, and twenty-one. These finite checks support the code; the preceding proofs establish the general claims.

10.2 Baselines that retain the elementary reductions

For optimization with fixed weights, the full disaggregation uses only positive-weight state flows, block balances $Af^j = \lambda_j b$, and native variable bounds $0 \leq f^j \leq \lambda_j u$. It eliminates x, z by substitution and adds the constant $c_y^\top \bar{y}$ to the objective value. In additional original-coordinate rows, the fixed y

contribution moves to the right-hand side. This is one joint LP and remains valid with the aggregate budget below. The global-merger baseline first groups labels observed nowhere into one default, then applies the same construction to its positive-weight states. With free weights, both interfaces retain every original y variable and its objective and constraint contributions. Both use sparse block and Kronecker assembly. They use no graph blocks, cycle coordinates, or observation rank elimination. Initial compression and observed elimination use the two variants of the general implementation, including their preprocessing costs.

For point membership, x, y, z are data and only state flows are LP variables. The historical full baseline already had this property. The strengthened baseline removes exactly zero-weight states, checks their observations, and assembles state balances and aggregate equations as sparse block matrices. It accepts solver statuses zero and two as numerical feasibility and infeasibility; other statuses are failures, not successful rejections.

An additional elementary baseline applies when precisely two states have positive weights λ, μ , with $\lambda + \mu = 1$. After checking the original aggregate flow, write the state flows as f and $x - f$. The complete remaining feasibility system is

$$Af = \lambda b, \quad \max\{0, x - \mu u\} \leq f \leq \min\{\lambda u, x\}. \quad (113)$$

An observation in the first state fixes $f_e = z_{ej}$; an observation in the second fixes $f_e = x_e - z_{ej}$. Intersect these fixed values with both bounds and with each other. The second state's balance follows from $Ax = b$, so this one-flow network LP is exact as a formulation. One positive state requires only a direct check. In the numerical implementation the aggregate balance precheck uses tolerance 10^{-9} ; its LP answer is still numerical evidence. Exact input arithmetic identifies zero states and checks the intersected bounds.

Fixed-weight linear optimization without additional coupling is also simpler than a joint LP suggests. If c_e^j is the product-objective coefficient for (e, j) , or zero when absent, and $c^0 = 0$, then disaggregation gives

$$\min_{(x, \bar{y}, z) \in H} (c_x^\top x + c_y^\top \bar{y} + c_z^\top z) = c_y^\top \bar{y} + \sum_{j \in \{0, \dots, m\}} \lambda_j \min_{v \in P} (c_x + c^j)^\top v. \quad (114)$$

This follows by independently minimizing each positive-weight state flow; zero-weight terms vanish. Unobserved labels have identical costs and share one solve. We include this sequential network-LP baseline. With free weights and no side constraints, a simplex vertex suffices, so such stand-alone objectives should not be presented as intrinsically difficult coupled problems.

10.3 Instances and measurement protocol

All reported data are synthetic. The code uses SciPy [19] and its bundled HiGHS LP solver [11], with default `highs` options. The recorded versions are Python 3.12.14, NumPy 2.5.2, and SciPy 1.18.1, on x86-64 Linux under WSL2 with 36 visible logical CPUs. The bundled HiGHS version is 1.12.0. No affinity, host isolation, persistent model, or solver warm-start advantage is claimed. Each case has one recorded warmup followed by five runs with method order rotated. Each run rebuilds its model. Build and solver times are recorded separately, together with total times and all minimum/median/maximum values. Component medians need not add to the total median. Independent certificate audits occur outside the timed components and have their own recorded cost. Membership-only and decomposition-inclusive exact queries are separate workloads, not timings inferred by subtraction.

The principal sparse optimization instance has four articulation-linked blocks, each containing five internally disjoint paths of length two, with a bridge between successive blocks. It has 43 arcs

Table 2: The 43-arc, 128-label optimization case. Times are milliseconds; total reports median [minimum,maximum] over five runs. The network baseline reuses one matrix for sequential solves. Repeated-cut build/solve columns exclude its separately recorded separator and reconstruction work, which the total includes.

Method	Vars.	Rows	Nonzeros	Total	Build	Solve
Full	5547	3612	11094	19.05 [17.81,23.05]	1.07	18.12
Global merge	516	336	1032	4.25 [3.41,5.01]	0.70	3.18
Initial compression	255	225	936	4.82 [4.70,5.62]	2.48	2.31
Observed elimination	222	192	839	6.45 [5.52,9.31]	4.11	2.47
Independent networks	43	28	86	22.02 [19.39,28.09]	0.37	21.64
Repeated cuts	207	179	630	292.22 [280.98,330.97]	14.78	100.57

and 128 explicit labels. Capacities are integers from two to eight, orientations vary, and balances are induced by the half-capacity reference flow. Each block observes three labels on three arcs each. The graph and label-generation seeds are fixed in the accompanying generator; the primary seed is 33, and objective seed 1033 gives independent normal costs with standard deviation 0.15 on x , zero on y , and standard deviation one on observed z . Weights are fixed to $9/(10m)$.

The membership family uses four six-path blocks of length two, seed 41, 51 arcs, 36 observations, and $m = 16, 128, 1024$. Each explicit state has weight $1/(2m)$, leaving residual weight $1/2$. In block B , state j perturbs path $(j + B) \bmod 6$ and the next path by opposite amounts $(j \bmod 3 - 1)/4$, with path orientation signs applied. The aggregate and observations are formed from this exact mixture around the half-capacity reference. These are feasible membership queries, not random rejection tests.

Two controls separate the mechanisms. First, an all-labels-observed instance has 16 three-path blocks, four disjoint labels per block, and one observation on the first arc of each path for each local label. All 64 labels occur somewhere, so global merging cannot help. It has 111 arcs and 192 observations; graph seed 73 and objective seed 1073 are fixed, with nonzero y costs and weights $9/640$. Second, on the original 43-arc instance, choose the nonnegative resource indicator $d_e = 1$ where an unconstrained optimum has x_e above the reference value, and zero elsewhere. Put

$$d^\top x \leq B, \quad B = \frac{1}{2}(d^\top x^{\text{opt}} + d^\top x^{\text{ref}}).$$

The exact reconstructed unconstrained point violates this row, while the reference graph point satisfies it. Every formulation receives the same aggregate row. This compares equivalent relaxations $H \cap \{d^\top x \leq B\}$; it does not assert that convexification commutes with adding a budget. The independent-state baseline (114) is inapplicable to this coupled control.

Finally the flat family uses G_L for $L = 8, 32, 128, 512$, observing $(a_i, 1)$ and $(a_i, 2)$ alternately. Boundary weights are $(1/2, 1/2)$; interior weights are $(1/3, 1/3)$, so the latter has three positive states. Feasible queries have $x_{a_i} = x_{b_i} = 1/4$, $x_h = 1/2$. Writing $g = i - 1$, the boundary observed value is $1/8 + (-1)^g(g \bmod 3 - 1)/16$; the interior value is $1/12 + (g \bmod 3 - 1)/48$. Branch profiles $1/4, 1/4$ in the boundary case and $1/6, 1/6, 1/6$ in the interior case give feasible completions: in the latter, split the remaining a_i aggregate equally between the other two states. Infeasible queries use $x_{a_i} = 3/10$, $x_{b_i} = 1/5$, $x_h = 1/2$, and every observed product $3/10$. All separate McCormick inequalities hold, but the two observed profiles would sum to at least $3/5 > 1/2$.

10.4 Size savings and runtime tradeoffs

Table 2 shows that global merging accounts for much of the full-versus-compressed difference on the original sparse instance. Only 11 of its 128 labels are observed anywhere: the full formulation’s

Table 3: Feasible membership on the 51-arc, 36-observation family: median construction-plus-query milliseconds. The exact-separator column is membership without requested decomposition; the raw data separately record constructive queries and all timing ranges.

Labels	Full	Global merge	Initial	Eliminated	Exact separator
16	7.42	5.35	6.21	7.65	5.93
128	44.65	5.01	5.54	7.14	5.23
1024	1580.67	9.29	16.16	16.25	13.68

Table 4: Two controls: median total optimization milliseconds. Independent network optimization is inapplicable with the aggregate budget. Repeated cutting was measured on the original and budgeted instance, not on the all-labels-observed control.

Case	Full	Global	Initial	Eliminated	Networks	Cuts
Aggregate budget	32.80	4.33	5.84	6.11	–	273.35
All labels observed	30.36	30.38	10.16	18.39	129.00	–

5,547 variables drop to 516 under global merging, then to 255 under initial compression and 222 under observed elimination. Elimination removes 33 of 48 auxiliary coordinates. It saves rows and nonzeros but is slower overall than initial compression in this run: its extra symbolic work is not offset by a solver saving. The globally merged and initially compressed LPs have median total times of 4.25 and 4.82 milliseconds, respectively; global merging has the lower median despite its larger model, although the five-run timing ranges overlap. Comparing only with the full formulation would overstate the specifically graph-local benefit.

The McCormick optimum is -7.287862 , compared with hull optimum -6.981917 . The repeated single-cut loop requires 42 cuts and is slower than the compact LPs. Its final decomposition is checked exactly, and each warmup cut receives an independent numerical global support audit. The independent-network baseline also attains the same objective, but multiple solver calls have overhead; algebraic separability does not guarantee the fastest implementation.

Table 3 gives the analogous membership comparison. At 1,024 labels, global merging is faster than either general compressed variant and the exact separator on these instances. The full LP’s much larger positive-state system is unnecessary when almost all labels are unobserved. The exact separator supplies a rational constructive certificate when requested, which is a different output from numerical LP feasibility.

The all-labels-observed control in table 4 isolates a benefit that global merging cannot provide. Both full and global formulations have 7,215 variables; initial compression has 495, and observed elimination has 367 with no auxiliary variables. Initial compression is substantially faster here. The aggregate-budget control retains the same size ordering and has objective -6.538573 . Global merging is also faster than initial compression on that control, with median totals of 4.33 and 5.84 milliseconds. Both remain faster than repeated cutting on this coupled relaxation. The budget and the local-label control are small targeted comparisons, not evidence about industrial instances.

The stronger baselines change the long-chain interpretation (table 5). On the 512-gadget boundary face, both the positive-state block LP and the one-flow LP are faster than the reduced exact oracle. Thus the earlier construction-inclusive advantage over a Python row-by-row full builder that retained the zero residual state does not survive these elementary improvements. The old builder already treated original coordinates as data; the changes are zero-state removal and assembly, followed by the one-flow substitution. At that boundary point, the positive-state LP has 2,050

Table 5: Flat-chain membership: median construction-plus-query milliseconds. Boundary weights are (1/2, 1/2); interior weights are (1/3, 1/3). Full retains only positive states. The one-flow reduction applies only with two positive states. Exact columns exclude optional decomposition; all ranges and constructive workloads are retained in the data.

L	Weights	Point	Full	One-flow LP	Unreduced exact	Reduced exact
8	Boundary	In	2.09	1.60	0.44	0.33
8	Boundary	Out	1.89	1.36	0.49	0.37
8	Interior	In	2.53	–	0.46	0.31
8	Interior	Out	2.90	–	0.79	0.51
32	Boundary	In	3.09	2.12	1.38	0.97
32	Boundary	Out	3.65	1.83	1.53	1.07
32	Interior	In	3.46	–	1.45	1.00
32	Interior	Out	2.48	–	1.32	1.04
128	Boundary	In	4.95	4.41	5.06	3.85
128	Boundary	Out	5.63	4.57	6.23	4.03
128	Interior	In	10.39	–	4.53	3.92
128	Interior	Out	4.12	–	5.34	4.11
512	Boundary	In	14.36	12.90	18.89	16.93
512	Boundary	Out	10.62	9.31	20.37	15.74
512	Interior	In	41.04	–	27.16	23.12
512	Interior	Out	39.18	–	51.05	41.23

variables and the one-flow LP has 1,025; the interior three-state LP has 3,075. On feasible interior queries the reduced oracle is faster at the largest length. Infeasible interior queries have similar medians and wide ranges: the full LP ranges from 16.08 to 45.27 milliseconds and the reduced oracle from 21.02 to 58.95 milliseconds. Five measurements on a shared host do not support a universal speed ordering.

The two-label oracle has no circuit-library setup. For comparison, the retained unreduced three-coordinate library takes 31.05 milliseconds to construct in the recorded cold call. The new three-observed-label library takes 10.78 milliseconds, with another 9.15 milliseconds for its inverse bases; these are separate from the two-label workload. Initial and observed-eliminated general models were remeasured on the eight flat cases at lengths eight and 32. They do not improve the total times over the dedicated exact oracle on those cases; their complete component and size data are retained. The earlier full-length compressed comparison is preserved as historical evidence, without using it to lead the current performance conclusions.

Rows in the tables exclude variable bounds. Nonzero and byte counts refer to input sparse matrices, not peak memory, solver factors, or Python rational objects. These measurements establish the behavior of the supplied code on the stated inputs. They neither compare optimized native implementations nor measure persistent callbacks in a larger optimizer.

10.5 A two-supplier service-transportation interpretation

A concrete modeling use is a transportation network with two suppliers s_1, s_2 and demand nodes $v_1, \dots, v_k$, with arcs from both suppliers to every demand node. Let supplies be S_1, S_2 , demands D_i , and arc capacities $u_{\ell i}$. The flow equations are

$$\sum_i x_{\ell i} = S_\ell \quad (\ell = 1, 2), \quad x_{1i} + x_{2i} = D_i \quad (i \in [k]), \quad 0 \leq x_{\ell i} \leq u_{\ell i}.$$

Suppose a service or contract allocation $y \in \Delta_m$ modifies the cost on selected route–service pairs. Its objective contribution has the form $\sum_{((\ell,i),j) \in \mathcal{O}} c_{\ell i,j} x_{\ell i} y_j$. Only pairs with a nonzero adjustment need product coordinates.

The underlying graph is a union of k internally disjoint length-two paths between the suppliers, even though the second arc on each path is traversed against its direction. It is therefore covered by [theorem 5.6](#). Demand conservation means that a circulation increase on the first supplier’s arc is the opposite change on the second supplier’s arc. The path coordinates retain these identities, while the state transportation problem enforces compatibility of all selected service products. With three demand nodes the theta formulas apply directly. For larger k , an exact transportation cut or compact EF is available; the number of observed service labels, their path coverage, and any repeated observations determine the useful reduction.

This interpretation uses the common balance and capacity domain assumed throughout the paper. An outer resource budget or linking constraint can be added to the exact component hull to strengthen a relaxation, as in the budget experiment. The resulting intersection is not generally the convex hull of the additionally constrained product graph. This is a modeling illustration; the experiments above use synthetic networks.

10.6 Reproduction

The accompanying repository includes the exact interfaces, independent verification programs, unchanged historical results, and a pre-upgrade source snapshot. The optimization comparisons were rerun after replacing the variable-weight LP baseline by the fixed-weight construction above. All three optimization cases retained their original objectives, weights, and additional rows; each method received a fresh audited warmup and five rotated runs. Membership and cold-library measurements were retained unchanged. The original measurements and sources are preserved in a hashed round-1 archive, and the correction record supplies a command for this optimization-only rerun. The complete measurement command, run from the repository root, is

```
PYTHONPATH=code python -m network_simplex_benchmarks.paper_stage06 \
  --output paper-network-simplex/verification/stage06-benchmarks.json
python paper-network-simplex/verification/stage06-tables.py
```

The table generator recomputes each reported minimum, median, and maximum from the raw five-run records and verifies the complete case grid. The versioned source, input definitions, objective vectors, additional rows, warmups, raw timing components, exact witness/cut audits, and numerical solver checks are identified by hashes in the validation manifest. The README files give the separate test and manuscript-build commands.

References

- [1] Mohammed Majthoub Almoghrabi, Martin Skutella, and Philipp Warode. Integer and unsplittable multiflows in series-parallel digraphs. *Mathematical Programming*, 2026. doi: 10.1007/s10107-026-02392-8. URL <https://link.springer.com/article/10.1007/s10107-026-02392-8>. Published online June 29, 2026.
- [2] Noga Alon and Van H. Vu. Anti-Hadamard matrices, coin weighing, threshold gates, and indecomposable hypergraphs. *Journal of Combinatorial Theory, Series A*, 79(1):133–160, 1997. doi: 10.1006/jcta.1997.2780. URL <https://web.math.princeton.edu/~nalon/PDFS/av1.pdf>.

- [3] Egon Balas. Disjunctive programming: Properties of the convex hull of feasible points. *Discrete Applied Mathematics*, 89(1–3):3–44, 1998. doi: 10.1016/S0166-218X(98)00136-X.
- [4] Danial Davarnia. *Convexification Techniques for Bilinear and Complementarity Constraints with Application to Network Interdiction*. PhD thesis, University of Florida, 2016. URL https://ufdcimages.uflib.ufl.edu/UF/E0/05/02/79/00001/DAVARNIA_D.pdf.
- [5] Danial Davarnia, Jean-Philippe P. Richard, and Mohit Tawarmalani. Simultaneous convexification of bilinear functions over polytopes with application to network interdiction. *SIAM Journal on Optimization*, 27(3):1801–1833, 2017. doi: 10.1137/16M1066166.
- [6] Jesús A. De Loera and Shmuel Onn. All linear and integer programs are slim 3-way transportation programs. *SIAM Journal on Optimization*, 17(3):806–821, 2006. doi: 10.1137/040610623. URL <https://www.math.ucdavis.edu/~deloera/researchsummary/universalitytransportation.pdf>.
- [7] Jack Edmonds and Richard M. Karp. Theoretical improvements in algorithmic efficiency for network flow problems. *Journal of the ACM*, 19(2):248–264, 1972. doi: 10.1145/321694.321699. URL <https://web.eecs.umich.edu/~pettie/matching/Edmonds-Karp-network-flow.pdf>.
- [8] L. R. Ford, Jr. and D. R. Fulkerson. Maximal flow through a network. *Canadian Journal of Mathematics*, 8:399–404, 1956. doi: 10.4153/CJM-1956-045-5. URL https://www.cs.yale.edu/homes/lans/readings/routing/ford-max_flow-1956.pdf.
- [9] L. R. Ford, Jr. and D. R. Fulkerson. Solving the transportation problem. *Management Science*, 3(1):24–32, 1956. doi: 10.1287/mnsc.3.1.24.
- [10] Peter Gritzmann and Bernd Sturmfels. Minkowski addition of polytopes: Computational complexity and applications to Gröbner bases. *SIAM Journal on Discrete Mathematics*, 6(2): 246–269, 1993. doi: 10.1137/0406019.
- [11] Qi Huangfu and J. A. Julian Hall. Parallelizing the dual revised simplex method. *Mathematical Programming Computation*, 10(1):119–142, 2018. doi: 10.1007/s12532-017-0130-5. URL <https://highs.dev/>.
- [12] Erfan Khademnia and Danial Davarnia. Convexification of bilinear terms over network polytopes. *Mathematics of Operations Research*, 50(2):1019–1041, 2025. doi: 10.1287/moor.2023.0001. URL <https://par.nsf.gov/servlets/purl/10546393>.
- [13] Tamás Kis and Markó Horváth. Ideal, non-extended formulations for disjunctive constraints admitting a network representation. *Mathematical Programming*, 194:831–869, 2022. doi: 10.1007/s10107-021-01652-z. URL <https://link.springer.com/article/10.1007/s10107-021-01652-z>.
- [14] Leo Liberti and Constantinos C. Pantelides. An exact reformulation algorithm for large nonconvex NLPs involving bilinear terms. *Journal of Global Optimization*, 36:161–189, 2006. doi: 10.1007/s10898-006-9005-4.
- [15] Garth P. McCormick. Computability of global solutions to factorable nonconvex programs: Part I—convex underestimating problems. *Mathematical Programming*, 10:147–175, 1976. doi: 10.1007/BF01580665.

- [16] Michal Melamed and Shmuel Onn. Convex integer optimization by constantly many linear counterparts. *Linear Algebra and its Applications*, 447:88–109, 2014. doi: 10.1016/j.laa.2014.01.007. URL <https://arxiv.org/abs/1208.5639v1>. Open preprint (2012), by Shmuel Onn and Michal Rozenblit.
- [17] Shmuel Onn and Uriel G. Rothblum. Convex combinatorial optimization. *Discrete & Computational Geometry*, 32:549–566, 2004. doi: 10.1007/s00454-004-1138-y. URL <https://arxiv.org/abs/math/0309083>.
- [18] Shmuel Onn, Uriel G. Rothblum, and Yoav Tangir. Edge-directions of standard polyhedra with applications to network flows. *Journal of Global Optimization*, 33:109–122, 2005. doi: 10.1007/s10898-004-4313-z.
- [19] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, et al. SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17(3):261–272, 2020. doi: 10.1038/s41592-019-0686-2. URL <https://scipy.org/citing-scipy/>.